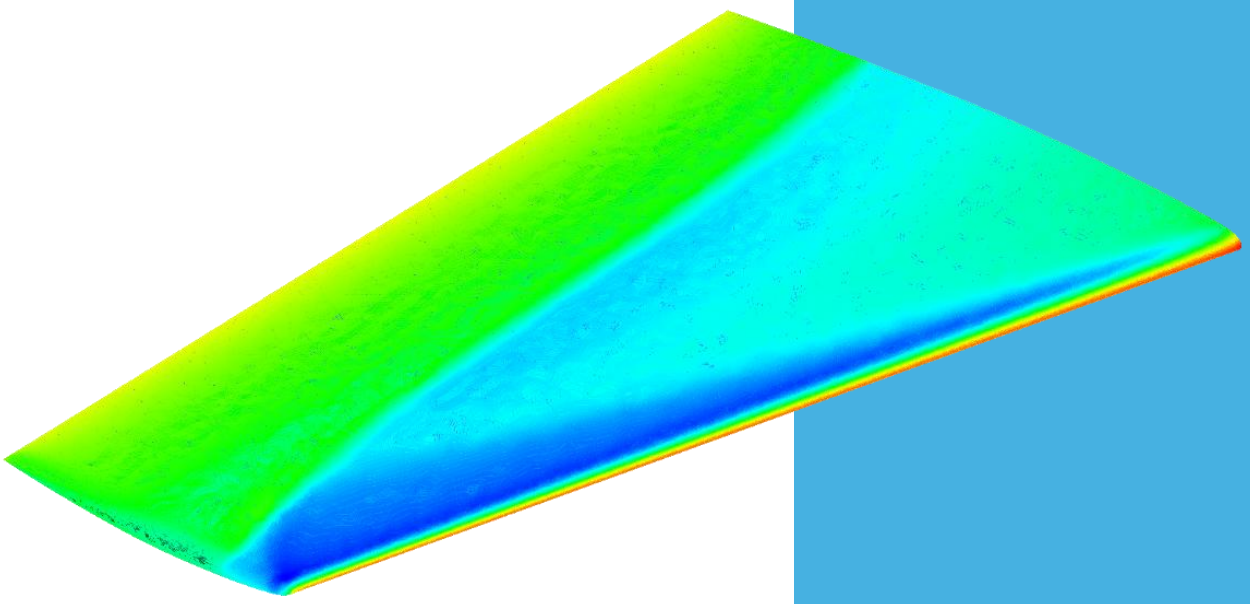
ONERA M6

Instructions on 3D Wing Simulation for Transonic Flow



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Computational Fluid Dynamics Simulation of a 3D Wing (ONERA M6) in ANSYS Fluent

Description: Structured Manual for Beginners on 3D CFD simulation

Keywords: *Aerodynamics, ONERA, CFD, Fluent, ANSYS, Transonic, Spalart-Allmaras*

I hope at this point, if you read through my 2D CFD simulation instructions, you are familiar with most of what CFD is. But nevertheless, we can revisit a few of them again.

Welcome to 3D CFD simulation in ANSYS Fluent. This is going to be grand!

(I hope this helps you in some way in your path to be an aerodynamicist – designing cool stuff that flies eh!?)

This instruction, following the same 'contextual base' of the 2D simulation, assumes for the most part that you are relatively new to computational physics domain, particularly in 3-dimensional CFD. The case-study we have in our hand is a transonic flow condition – on arguably the most citable design in transonic literature – NASA's ONERA M6 wing.

ANSYS Fluent* for Dummies who wants to do 3D transonic analysis, if you will.

(*Conditions apply – Although this is for a transonic flow, most of the things you learn will directly port to subsonic, and supersonic flow analysis – with a few caveats. What are the caveats? I am still learning don't ask me!)

The module will be divided into 4 parts –

- **Theory** – What Exactly is Computational Fluid Dynamics
- **Geometry Clean Up** – Opening your favorite CAD platform before we touch ANSYS
- **Fluid Domain and Meshing** – If I am being honest, this is 80% of any computational analysis. Meshing is as much as science, as it is art!
- **Fluent Solver** – Finally, the design is ready for the flow (to be set up)

Grab some coffee (the go to drink for aerodynamicists… I think), and feel free to skip the first part if you went thoroughly on the 2D CFD instruction.

Good Luck. You're gonna need it!

Part 1

THEORY: WHAT IS COMPUTATIONAL FLUID DYNAMICS?

So, you want to be (and think like) an aerodynamicist? You want to wow yourself (and later, anyone who dares to be drawn into your madness) with real world physics of how fluid (we will only discuss about air for now) acts around an object? You want to plot fluid mechanical coefficient like it's nobody's business?

Welcome to the sacred world of Computational Fluid Dynamics – where objects show you how good (or bad, no judgement here) they are in fluid loads, and where velocity vectors paint Bernoulli's equations and Navier-Stokes equations like Vincent van Gough.

Imagine a wing cutting through air. There will be various forces in and around the wing surface acting based on the classical mechanics principles (Sir Issac Newton). This, in turn, will result in forces like drag and lift – the bread and butter for our fluid dynamic analysis. But since there are a lot of components involved in such process, figuring out these key values (that will help you improve the aerodynamic performance) with formulas is a daunting task. Who has the time to do a few million calculations every time your object makes a slight adjustment in its position?

Introduce Computational Fluid Dynamics, the prodigal son of Finite Volume Method (or Finite Difference Method), a numerical tool that you've probably learned in college and wondered why am I even learning this?

CFD is a numerical method designed to solve physical problems by dividing a complex domain in smaller chunks (mesh). Under the hood, it is basically a differential equation solver that tackles the Navier-Stokes equations by discretizing (turning PDEs into algebra) and solving with iteration.

Here is a 'hack' version of what CFD is –

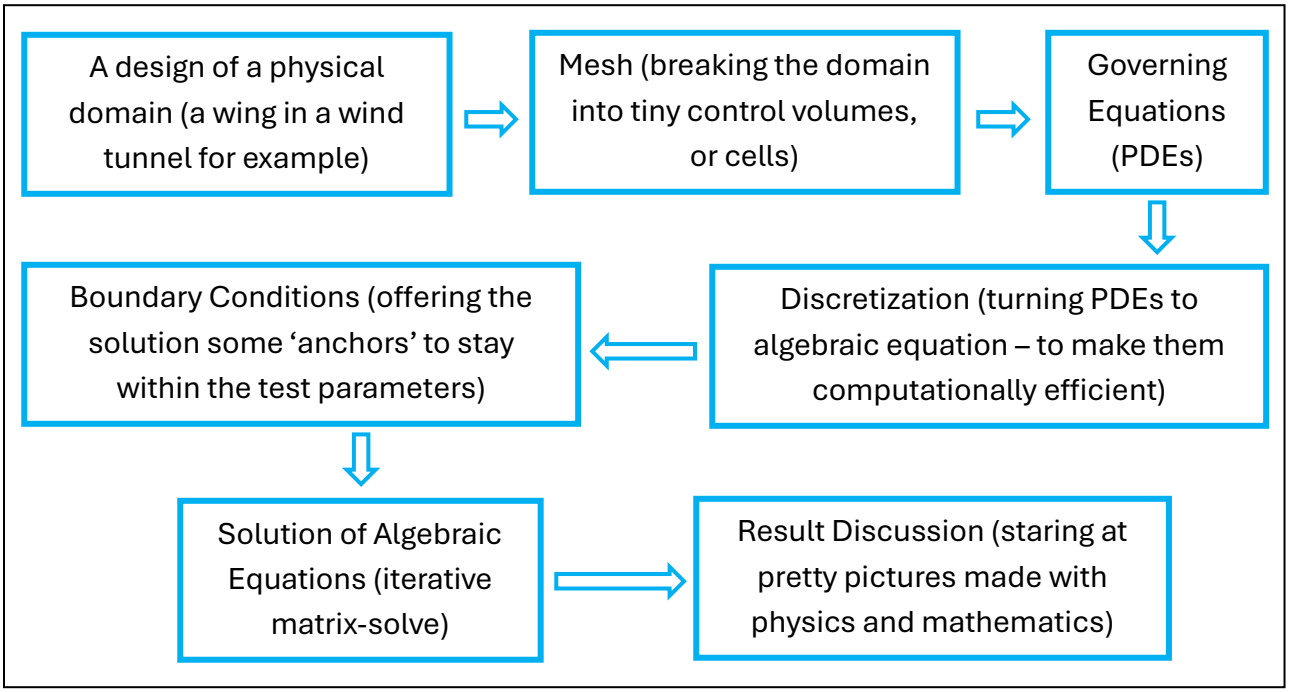


Figure 1.1: The 'hack' version overview of what CFD is

Congratulations. Now you know what CFD is better than your classmate who got a 4 in her fluid mechanics course but has a visible twitch the moment she sets foot in the uncharted domain of ANSYS.

Which is a completely useless skill to have, but you still have it.

And I am going to stop right here before it turns into a lecture. Let us see how exactly we can set up our own model for a complete simulation run.

As stated, this instruction will work with ONERA M6 wing, a canonical example of 3D wing design for transonic literature. The flow conditions are as stated –

Parameter	Value
Freestream Velocity	Mach 0.84
Angle of Attack	3.06 °
Characteristics Length	0.64607 m
Reynold's Number	14.6 million (from literature)

For the sake of scaling our simulation for parametric studies in future, the AoA setup is purely based on trigonometric function. However, you can rotate the wing itself before meshing, but that is not advised. Keep the wing steady, rotate the flow direction – is the proper way anyways.

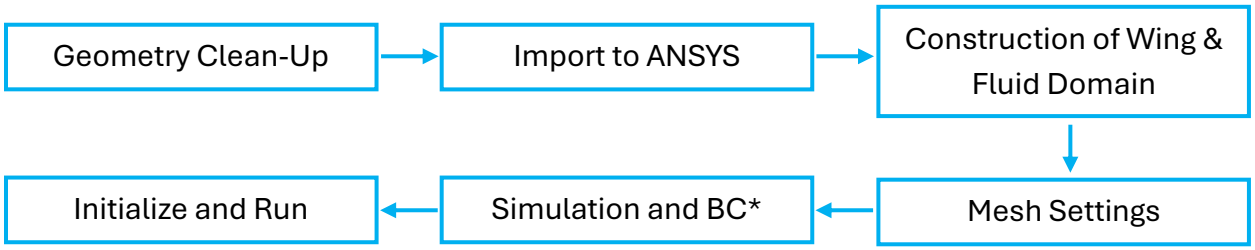
(You can use a calculator for it, or do some sort of coding 'trickery' for this part. Tech-savvy readers are encouraged to check out this python trick on the subject matter)

Part 2

GEOMETRY CLEAN-UP

Let the Fun Begin!

Before we start with the nitty-gritties, let's take a look at the overview of the process, and then we can start building our own simulation shenanigans from the ground up.



**BC – Boundary Conditions*

Please note, the result discussion is a separate entity, and is beyond the scope of this instruction. There is a separate section for results in my website, you may find it here.

Let us start, shall we?

First and foremost, we need to fetch the geometry. Download the CAD file for ONERA M6 wing from NASA turbulence modeling resource database here.

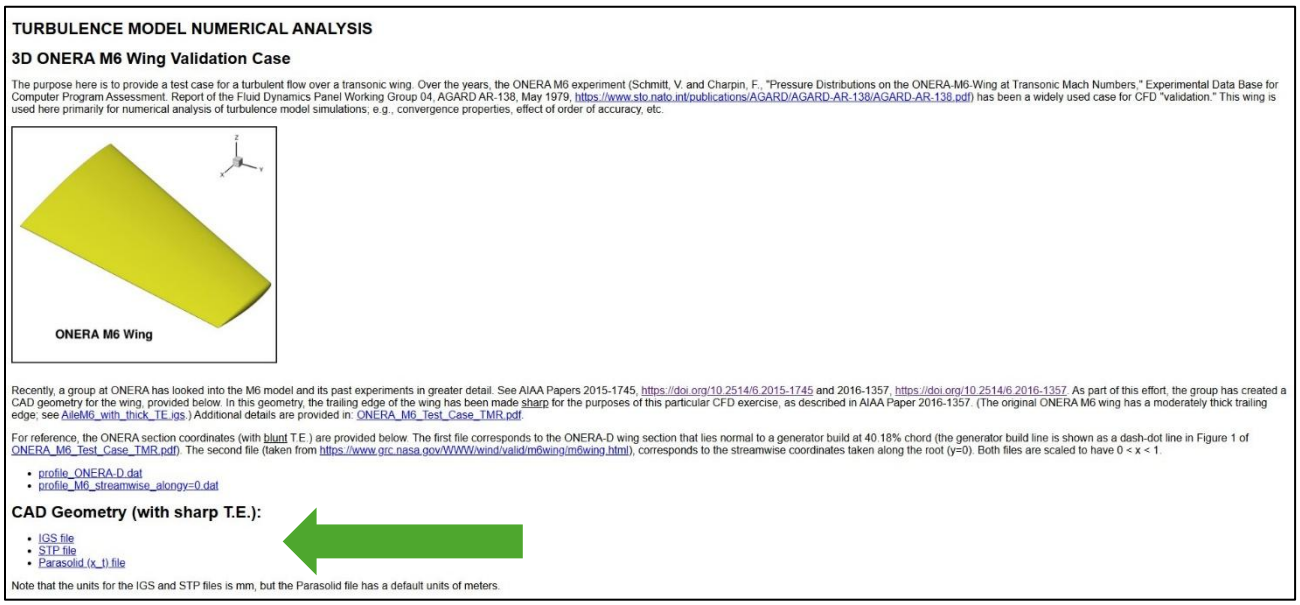


Figure 2.1: Turbulence Model Database Website

In this case, I am using the IGS file format from the database (**Figure 2.1**).

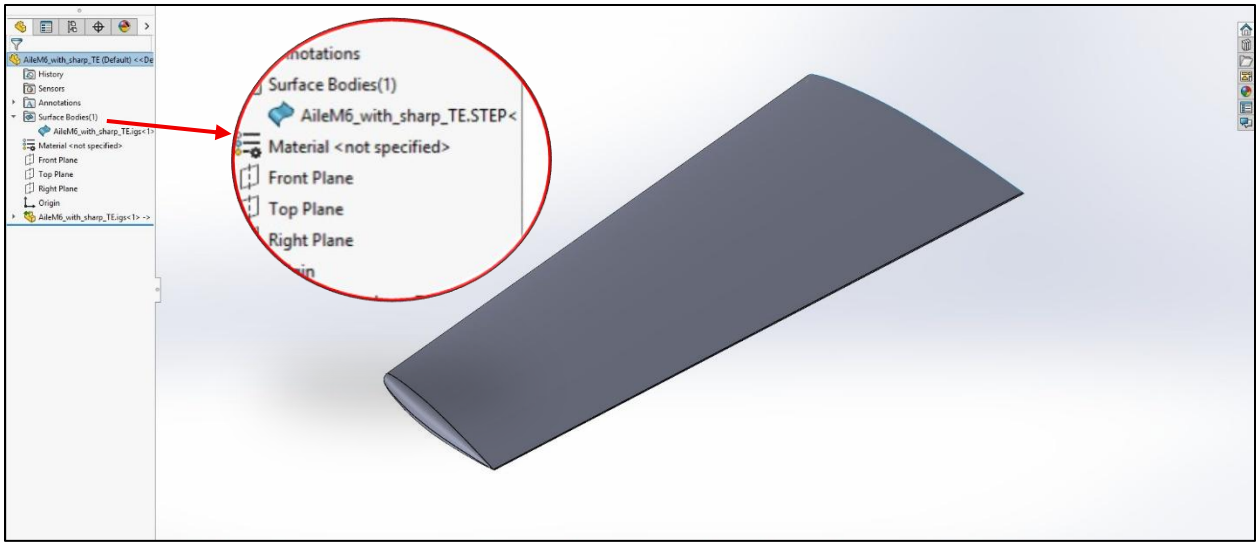


Figure 2.2: Downloaded CAD model in SolidWorks

Lucky for us, the default geometry in the website has already been 'cleaned-up'! So, I guess, there is no point in cleaning up the geometry in this case!

However, it's still a good practice to open the model in a CAD environment before we get to ANSYS. Sometimes, the geometry you may fetch will have some additional feature (say, a plate at the symmetry surface for example) baked into it. In that regard, it is advised to clean them up (removing any unnecessary feature, say through cutting those parts) to make sure you only have the wing. In the next section, all we need is a single, clean geometry that can be 'extracted' out from the fluid domain surrounding it.

To wit, it's best to make sure in the CAD environment that there is only 1 surface body (**Figure 2.2**). If all looks good, time to export it to a 'favorable' format. ANSYS can be very picky with CAD formats, and I am seeing consistent performance with *STEP AP214* format.

Once the export is done, we are ready for ANSYS.

Part 3

FLUID DOMAIN & MESHING

3.1 Fluid Domain Design

Yes, we can finally open ANSYS!

In Workbench, drag the *Fluid Flow (Fluent)* module onto the main project area. Right click on *Geometry*, and navigate to *Replace Geometry*. Browse to your file and import that shiny new .stp file we just exported from SolidWorks (or your choice of CAD software).

Once loaded, again right click on *Geometry*, and *Edit Geometry in DesignModeler…*

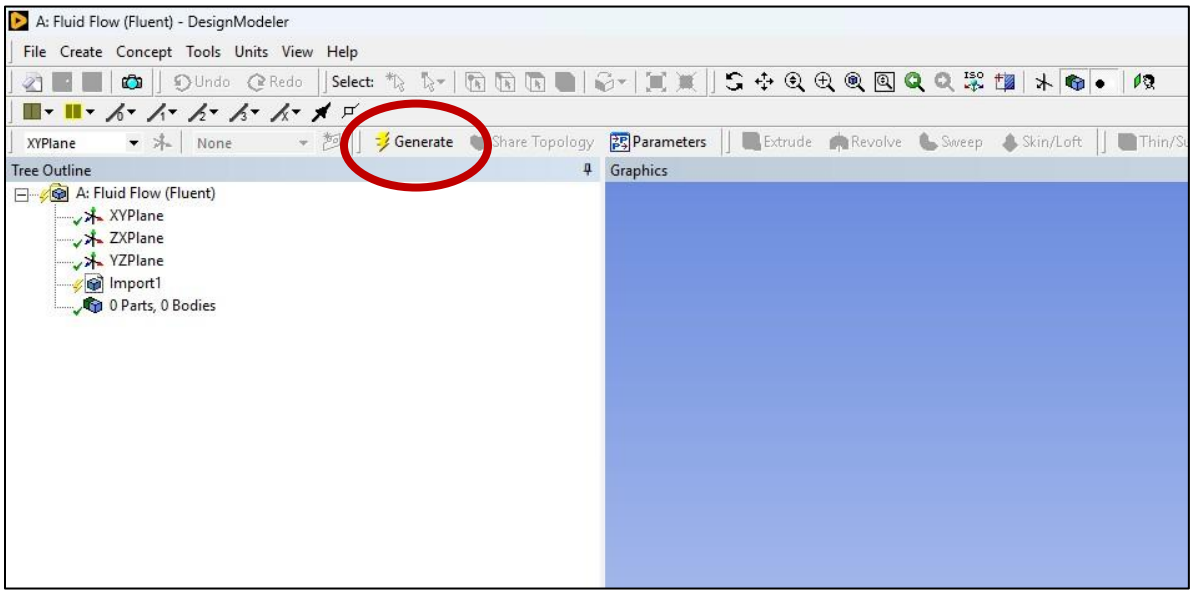


Figure 3.1: UI of Design Modeler

In DesignModeler, hit *Generate* – and you should see the wing model appearing in the viewport. Now, it's time to 'extract' the wing.

> *We will develop a fluid domain around the wing, and then subtract the wing from the domain. Essentially, we will be left with the fluid domain only, with a wing-shaped hole in the middle.*

Pay attention to the 'flushed' plane to the root of the wing. In our case, it's the ZX plane. From the feature tree, select the *ZXPlane* and click New Sketch (**Figure 3.2**).

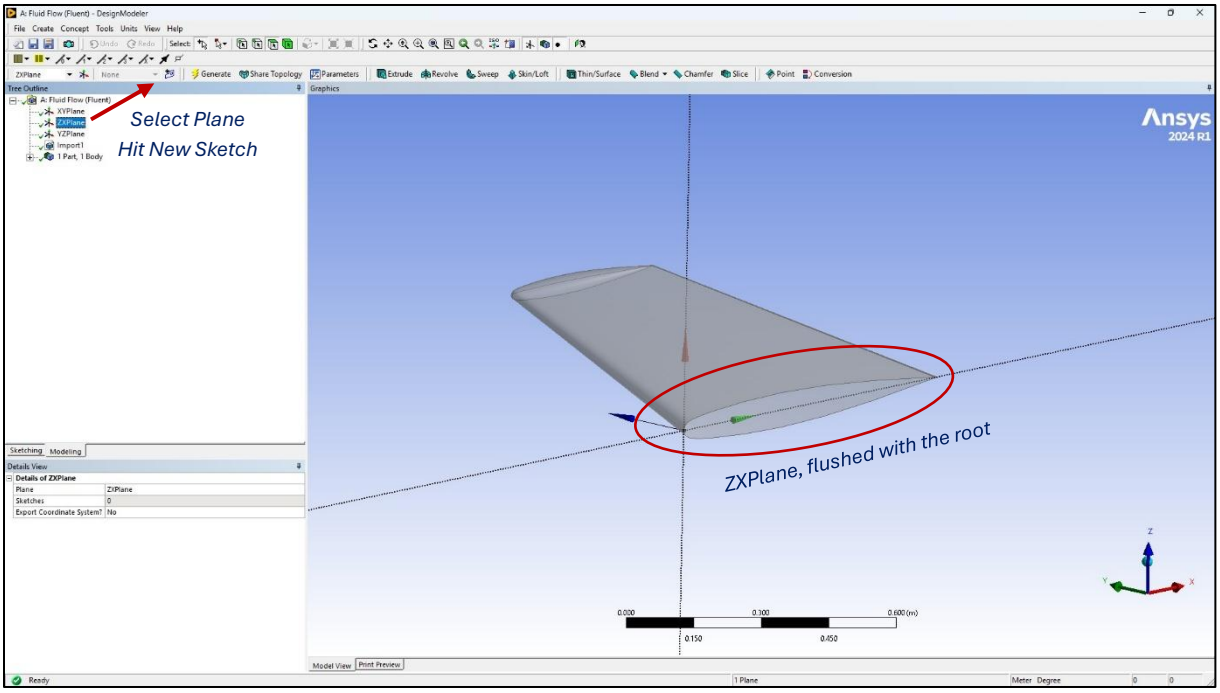


Figure 3.2: Plane Selection (based on the root chord plane) for sketching

Go to the *Sketching* tab, select *Circle* and click the origin in a way so that you see the letter P next to your pen (cursor). Once the circle is drawn, go to Dimensions in the Sketching tab and make sure General is selected. Put the value of 10m (d > 12r, as r = 810mm in our case) for the circle. This defines the radius of the farfield Riemann hemisphere. Understandably conservative, but it still offers decent accuracy within a manageable compute budget.

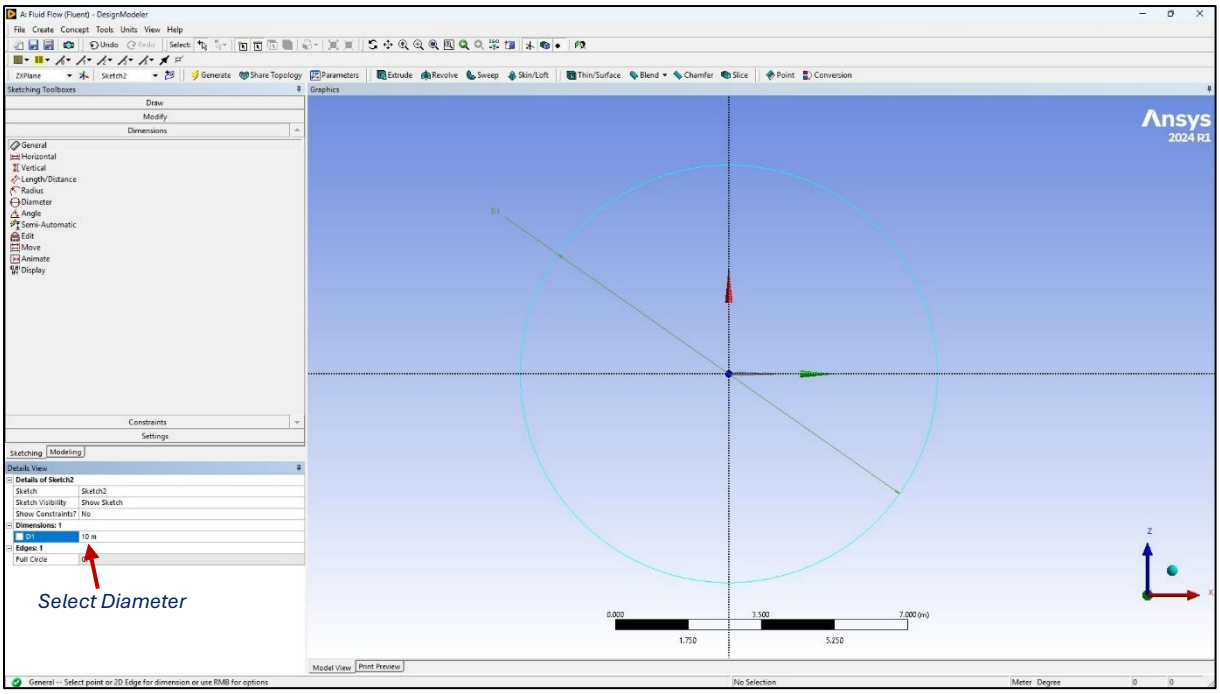


Figure 3.3: Selecting the dimension of the radius of Riemann hemisphere

Now, let's make it a semi-circle. Draw two lines from origin (in +Z and -Z directions), without touching the circle. Make sure you have the letter H and C appearing (**Figure 3.4**) as you draw the two lines. Go to the *Modify* section, and hit *Extend*. Both lines should touch the periphery of the circle. Use the *Trim* tool in Modify tab, and remove the left half of the circle (**Figure 3.5**).

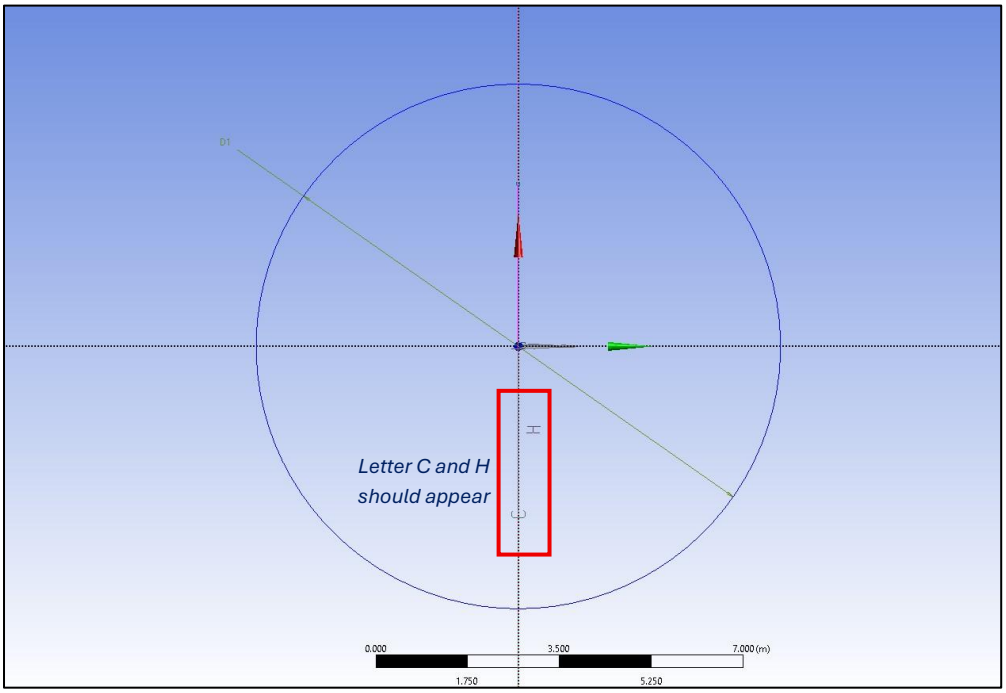


Figure 3.4: Drawing lines to dissect the circle

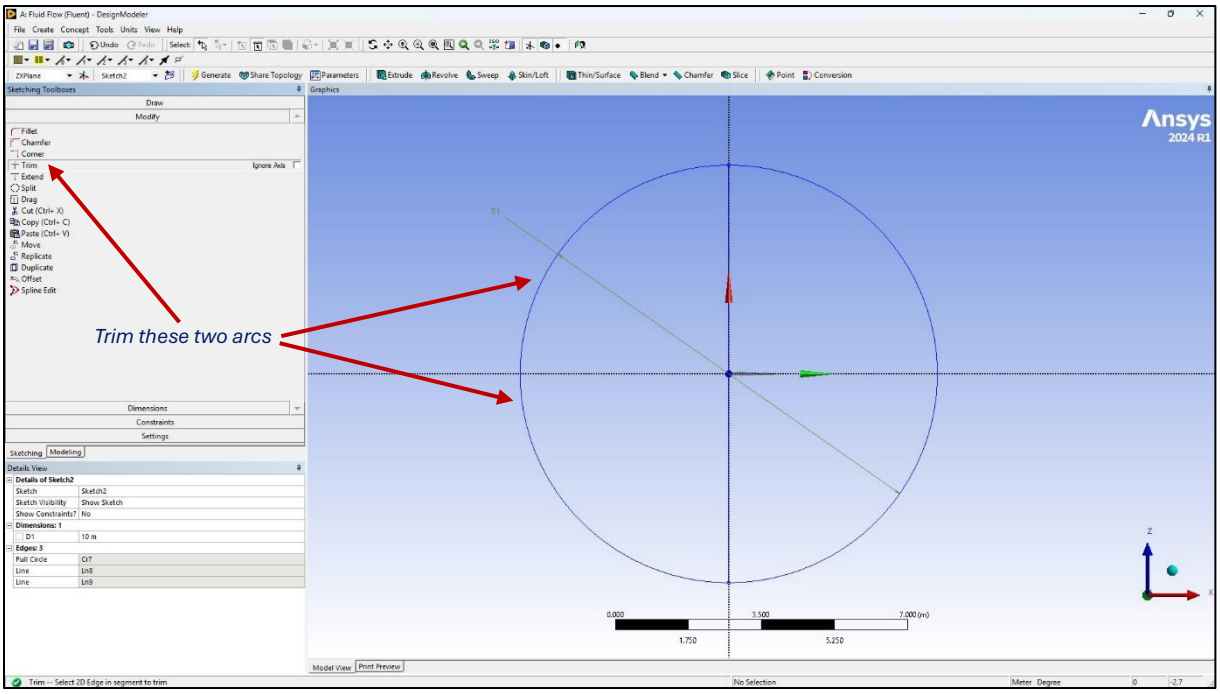


Figure 3.5: Trimming to make semi-circle

Time to make the hemisphere. Hit the Revolve feature on top of the viewport (**Figure 3.6**), and select the geometries as shown in that figure. Select the arc for geometry, and hit apply. Same for the line as the axis. If all looks good, hit Generate. And you should have a sphere model (**Figure 3.7**).

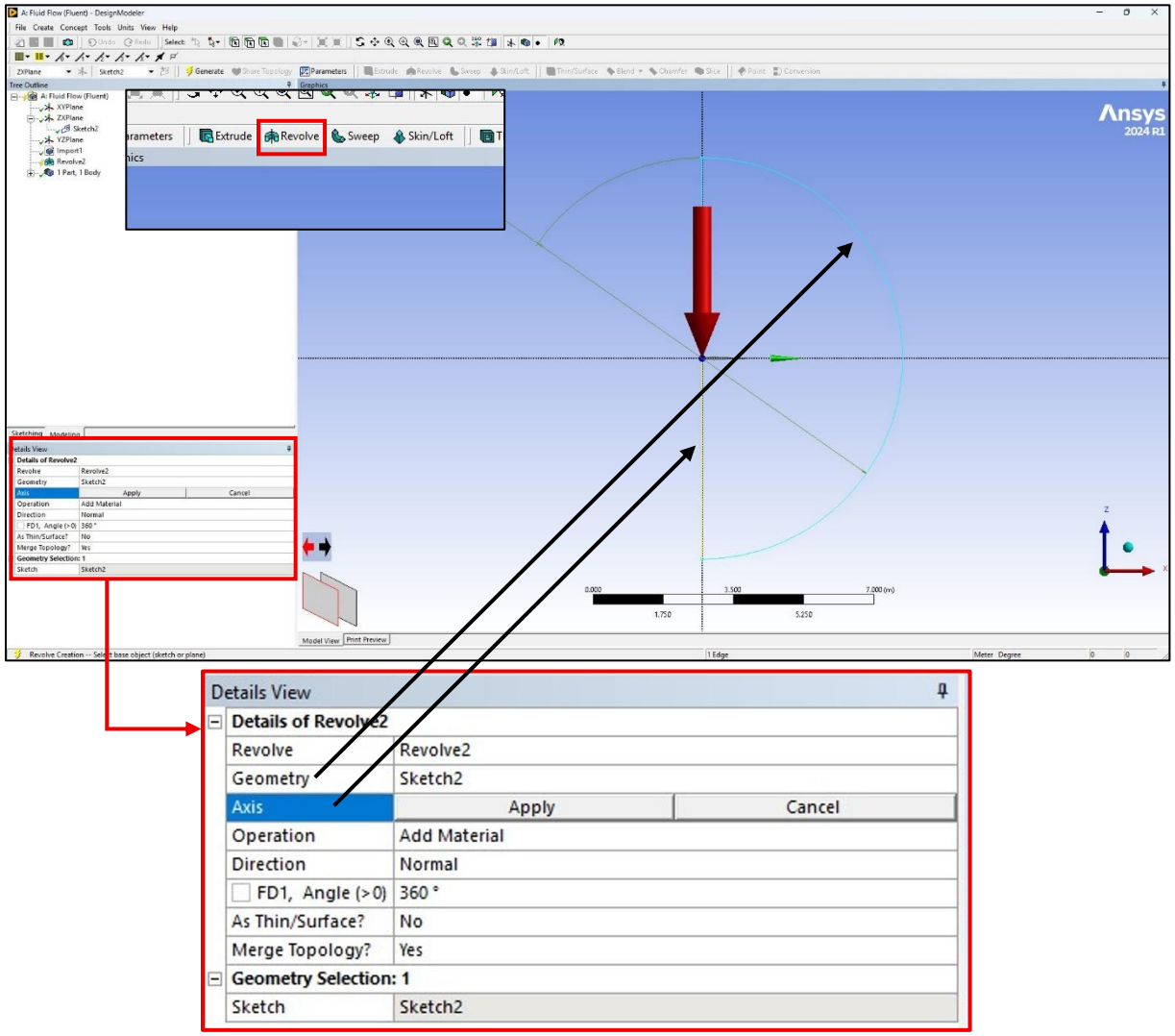


Figure 3.6: Using revolve to generate sphere from the semi-circle

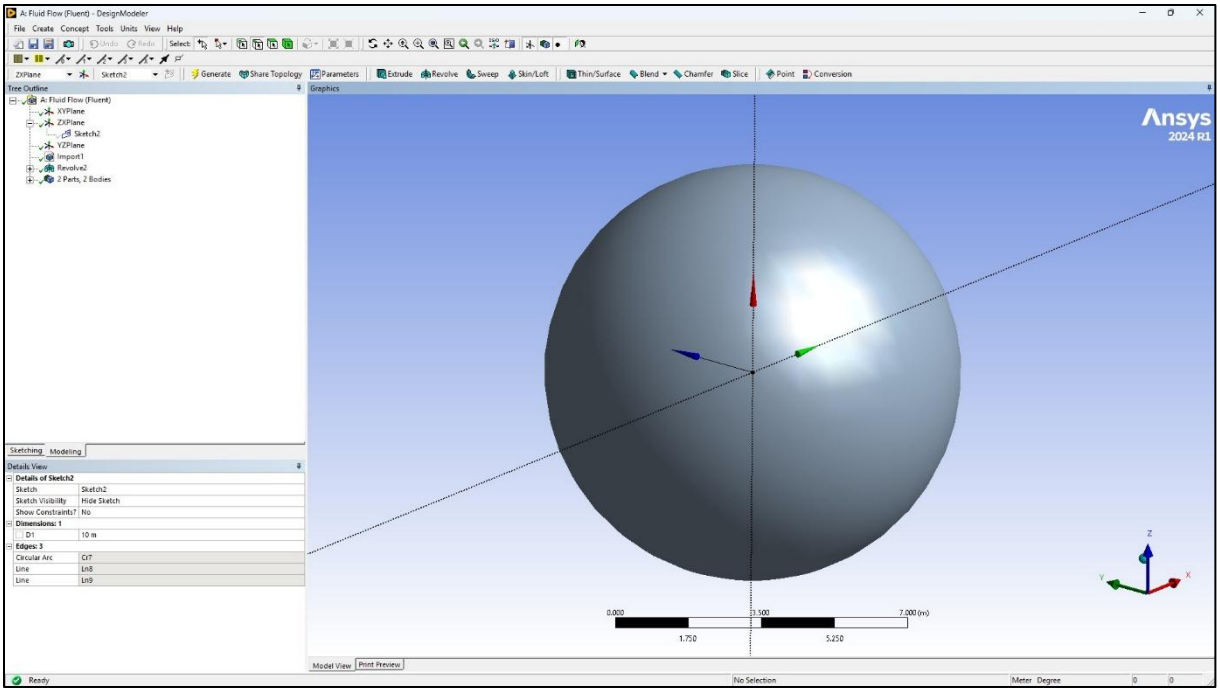


Figure 3.7: Sphere (duh!)

Select the ZXPlane (or the plane that was flushed with the root chord), select *Slice* from the same ribbon that has the Revolve feature we used earlier, click Apply in the Detail tab for *Base Plane*, and hit Generate. Now, the sphere is split between two halves. Suppress the part that does not contain the wing model (in our case, it is the one in the -Y direction) by selecting it first, right click and hit *Suppress Body*. You should have something like this –

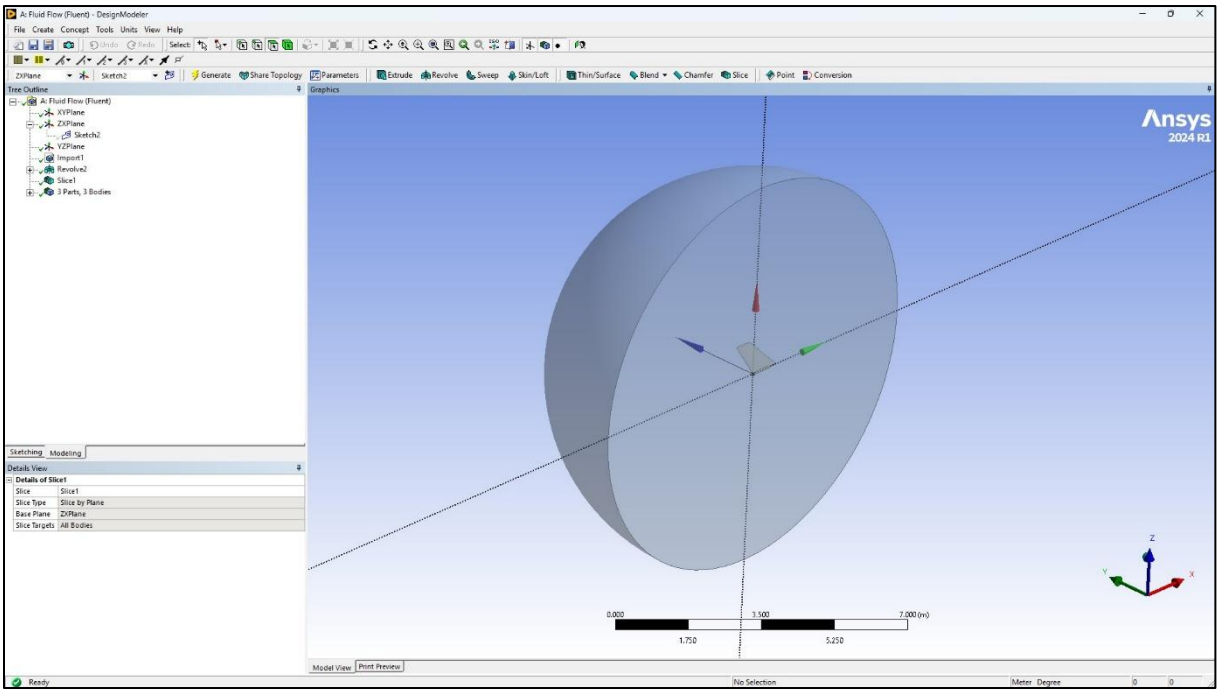


Figure 3.8: Hemisphere with the wing model

In the top ribbon, find the *Create* option, select *Boolean*, and select *Subtract* operation in the Details View tab. You will be prompted to select a target and tool body. For the target body, select the hemisphere, and select the wing for tool body. Please note, you need to use the 'layering' thing on the bottom left of the viewport to select the tool body (**Figure 3.9**). After selecting both the target and tool bodies, hit generate.

The wing has been 'extracted' from the fluid domain. Now, we have a working 'model' to run our simulation. But of course, there is more! This is the part where we define a few things for targeting finer mesh around specific region. Well, by few, I mean just one. We need to prepare a 'Body of Influence' around the wing – that will help us refining mesh where the fluid flow needs to be captured with a higher solver resolution. But before we do that, from the feature tree, suppress the solid body (that represents the wing), rename the other solid body as 'fluid_domain', and select *Fluid* for it (**Figure 3.10**). Hit Generate, and we are ready for Body of Influence design.

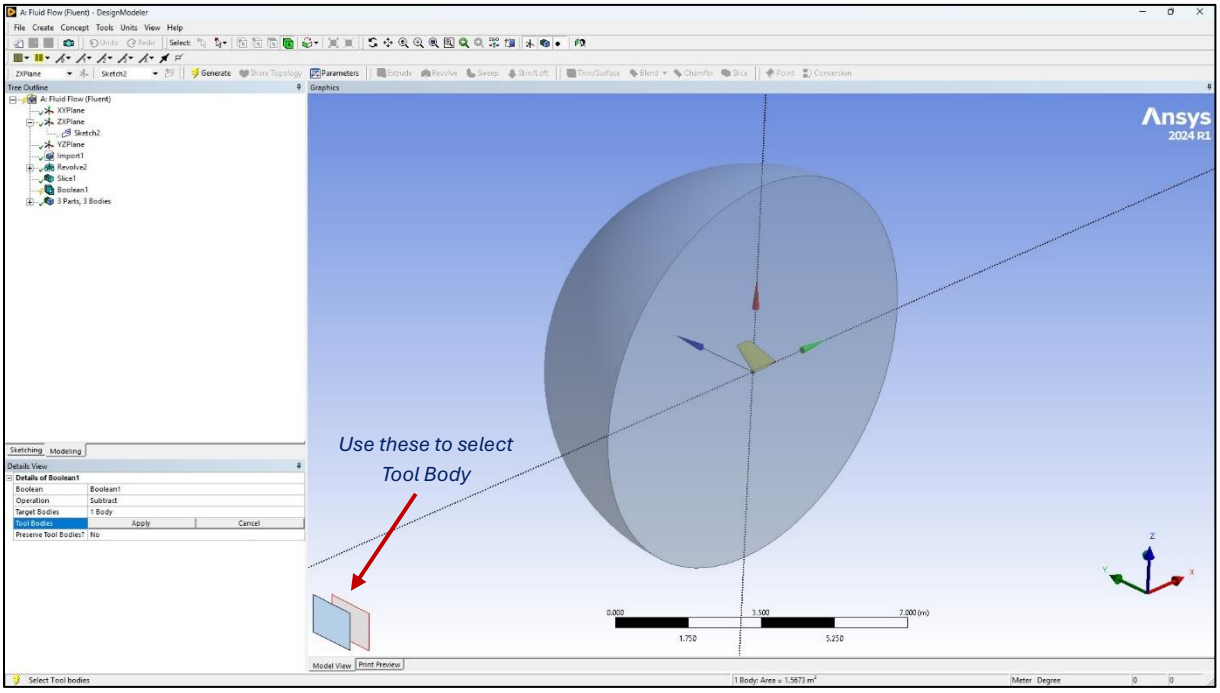
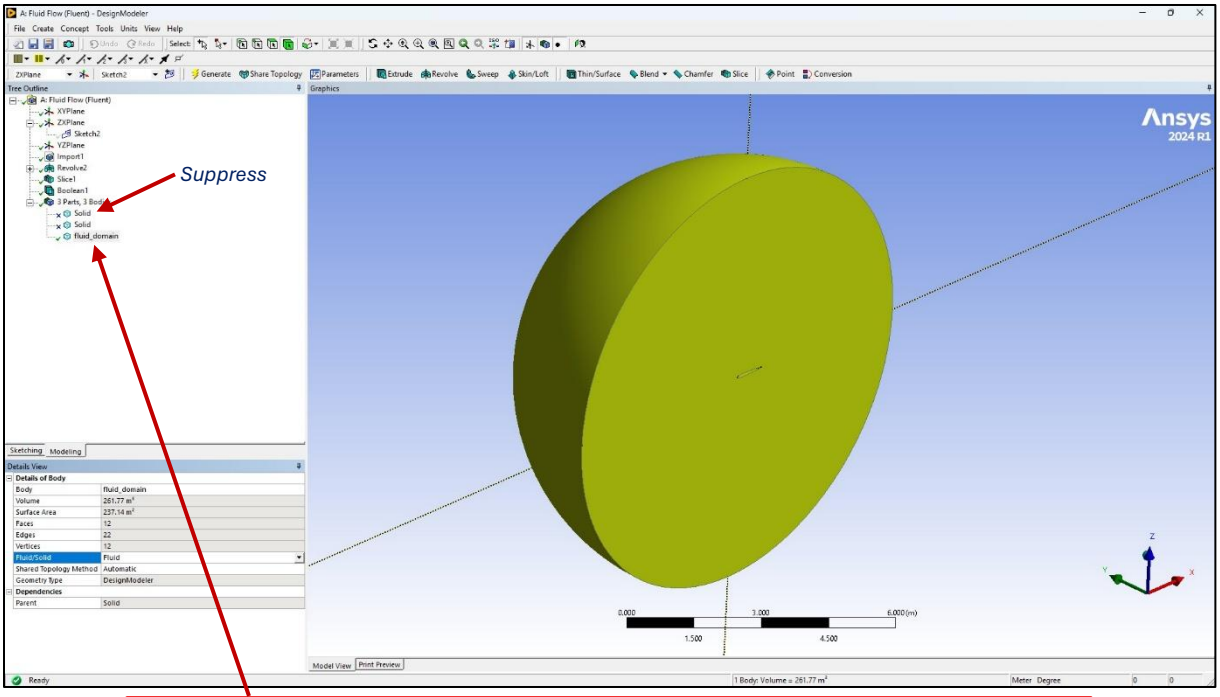


Figure 3.9: Boolean Subtract Operation



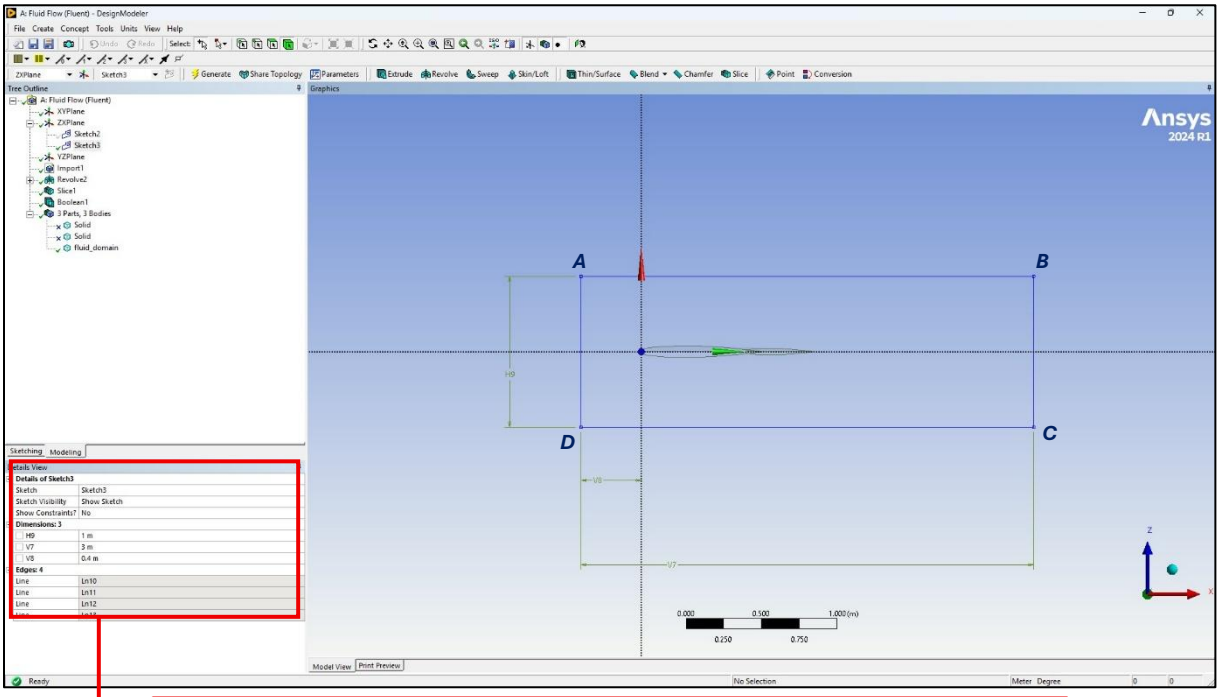
Details View	
Details of Body	
Body	fluid_domain
Volume	261.77 m^3
Surface Area	237.14 m^2
Faces	12
Edges	22
Vertices	12
Fluid/Solid	Fluid
Shared Topology Method	Automatic
Geometry Type	DesignModeler
Dependencies	
Parent	Solid

Figure 3.10: Fluid domain with a wing-shaped 'hole'

Select *ZXPlane*, click *New Sketch*, and draw a rectangle around the wing. The dimensions I used are arbitrary, but they may need to be revisited for your own geometries (and other cases) based on how you want to solve, what exactly you want to solve. For now, you can follow the dimensions shown (**Figure 3.11**) in this particular case. Also, use symmetry constraint from the sketch tab (select X-axis, then select the two sides parallel to X).

(For example – if there is a large shock signature over your wing, you may want to give more offset to the rectangle above the chord to capture more of that flow. As stated, this geometry will be used for refining mesh around the wing, it will make sense later on I promise!)

Important: Since our base model is oriented in a different way (Y-axis as depth rather than traditional Z-axis); the dimensions will be 'weird' in DesignModeler. For example, the length of the CD line is defined with a Vertical dimension tool in the Sketch tab, rather than horizontal. Similarly, DA follows a Horizontal dimension convention. No, you didn't learn geometry wrong in the school – it's just a weird quirk originated from the obscure orientation of the CAD model itself. However, you can use the General option for all the measurement – but I'd argue to use particular ones for more control; say the distance between AD and the Z-axis cannot be defined with General option.



Sketching | Modeling

Details View

Details of Sketch3	
Sketch	Sketch3
Sketch Visibility	Show Sketch
Show Constraints?	No
Dimensions: 3	
H9	1 m ***(DA)***
V7	3 m ***(CD)***
V8	0.4 m ***(distance between DA and Z-Axis)***
Edges: 4	
Line	Ln10
Line	Ln11
Line	Ln12
Line	Ln13

Figure 3.11: Dimensions of Body of Influence Sketch

Use the Extrude feature, hit generate, and we have our Body of Influence. Close DesignModeler, and save the project. Finally, we are ready for meshing!

3.2 Meshing

Before we start with minute details of targeting regions for meshing, it's always a good practice to Generate a base mesh. If the model has some issues, this is a good first-pass for it. In the meshing module, click the *Mesh* from the feature tree and just hit *Generate*. We don't need to worry about any sizing of method for now.

(Of course ours will, since the geometry is clean – but you'd be surprised how many of perfectly 'clean' looking CAD geometry of yours might fail this initial test, particularly when it comes to aerodynamic components with complex curvatures, and the underlying watertightness quality of the CAD model of the components.)

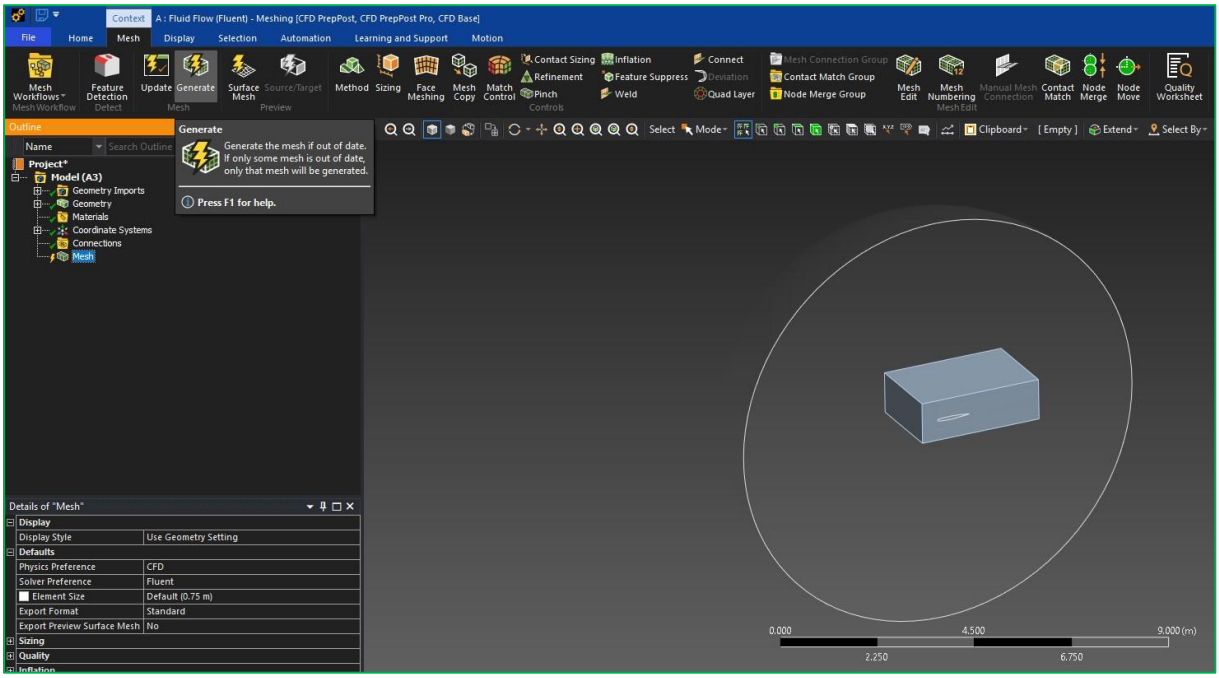


Figure 3.12: Generating base mesh on the geometry

After the base mesh is generated, we can start with 'engineering' the fluid domain with multiple regions, specific settings, all the bells and whistles. For an overview, the following refinement will be entertained for our mesh –

a. Body of Influence Sizing
b. Airfoil Edge Sizing
c. Spanwise Edge Sizing
d. Wing Surface Sizing
e. Inflation Layer (tied to y+ value estimation)

Once all of them are done, we define a few regions (farfield, symmetry) and the mesh will be ready to be ported to the solver.

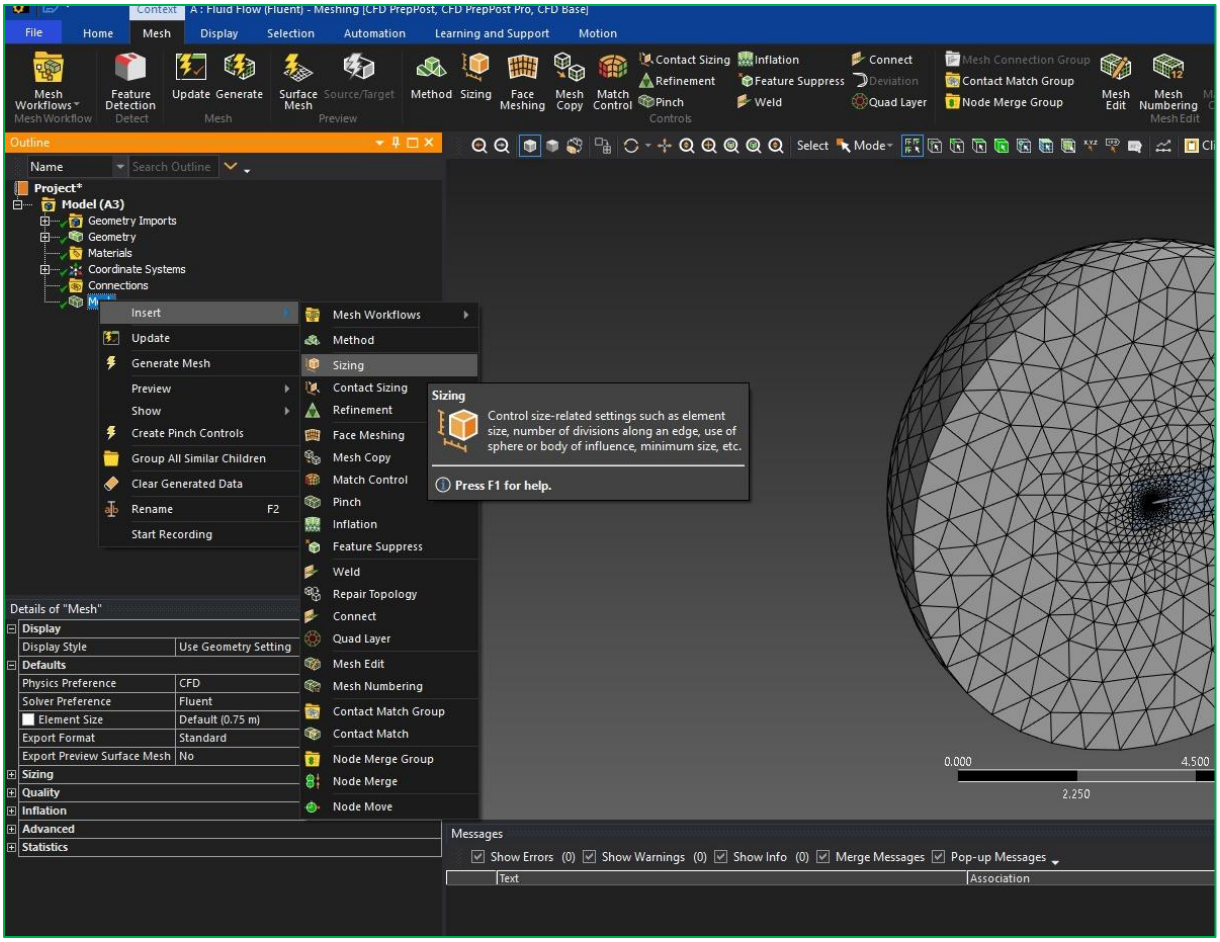


Figure 3.13: Sizing Selection for Mesh

Let's start with sizing, shall we? Just right click on *Mesh*, select *Insert* and navigate to *Sizing* (**Figure 3.13**). The process is same for all the sizing methods discussed below (except for Inflation, where you should go to Insert > *Inflation*).

3.2.a Body of Influence Sizing

Mesh > Insert > Sizing

With the *Body* selection tool (see the icon at the top ribbon, or Ctrl+B), select the fluid domain as *Scoping Method*, and select Body of Influence in *Type* (Figure 3.14). Select the solid body we made in the last stage of the fluid domain preparation, using the same 'layering' shenanigans at the left-bottom corner. An arbitrary *Element Size* of 0.04m has been selected, with the default Element Size of 0.15m (you can find the latter settings when you click Mesh on the feature tree).

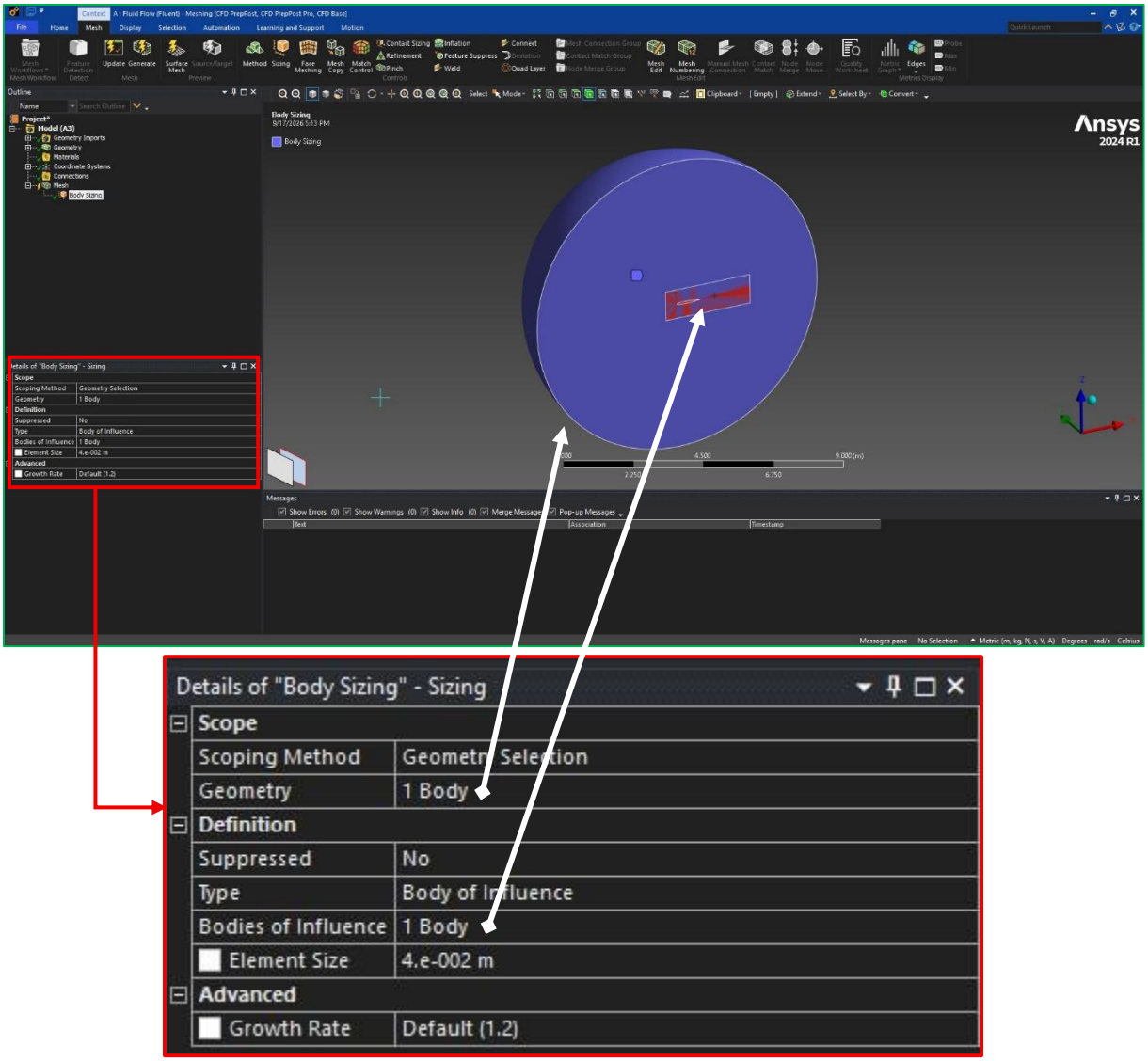


Figure 3.14: Body of Influence Mesh Settings

Now hit update in the Meshing tab, and wait for the mesh to finish. Technically, you can put all the sizing settings and generate the full mesh at once (which you eventually will do anyways), but it's a good practice to generate mesh after every new sizing method being added – that way we can diagnose a failed mesh the moment a failure point is introduced. And also, this is a good practice when we are still learning the process.

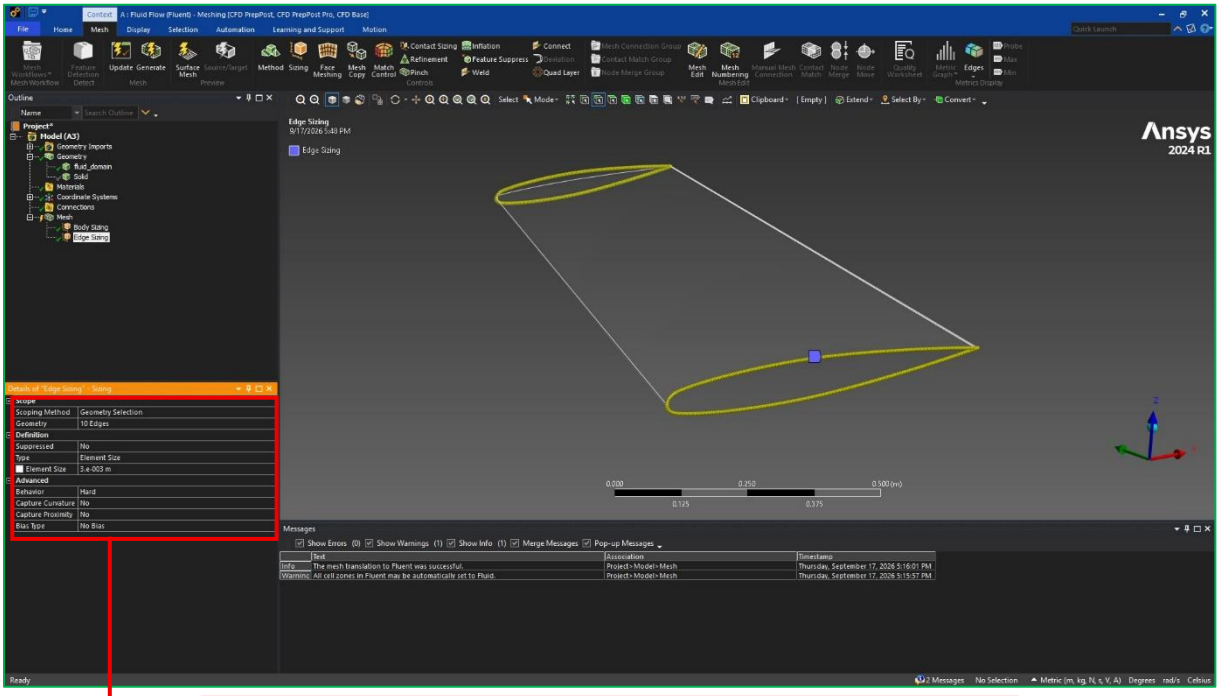
Before going to the next sizing section, we can now hide the solid body from feature tree. Hit the + sign beside *Geometry*, find the *Solid* part, right click and hit *Hide Body*. This makes things a bit easier down the line, especially when we need to select intricate surfaces and face features on the fluid domain.

Note – Do not suppress the body, just hide it.

3.2.b Airfoil Edge Sizing

Mesh > Insert > Sizing

With the *Edge* selection tool (see the icon at the top ribbon, or Ctrl+E), select the following edges carefully. You should have a total of 10 edges selected. There are two edges on the root, and 4 edges on the tip – that can easily be missed. Holding Ctrl, you should select all these edges. Once selected, use an arbitrary Element Size of 0.003m – while keeping all the other settings as is (**Figure 3.15**). Hit update.



Scope	
Scoping Method	Geometry Selection
Geometry	10 Edges
Definition	
Suppressed	No
Type	Element Size
Element Size	3.e-003 m
Advanced	
Behavior	Hard
Capture Curvature	No
Capture Proximity	No
Bias Type	No Bias

Figure 3.15: Airfoil Edge Sizing Mesh Settings

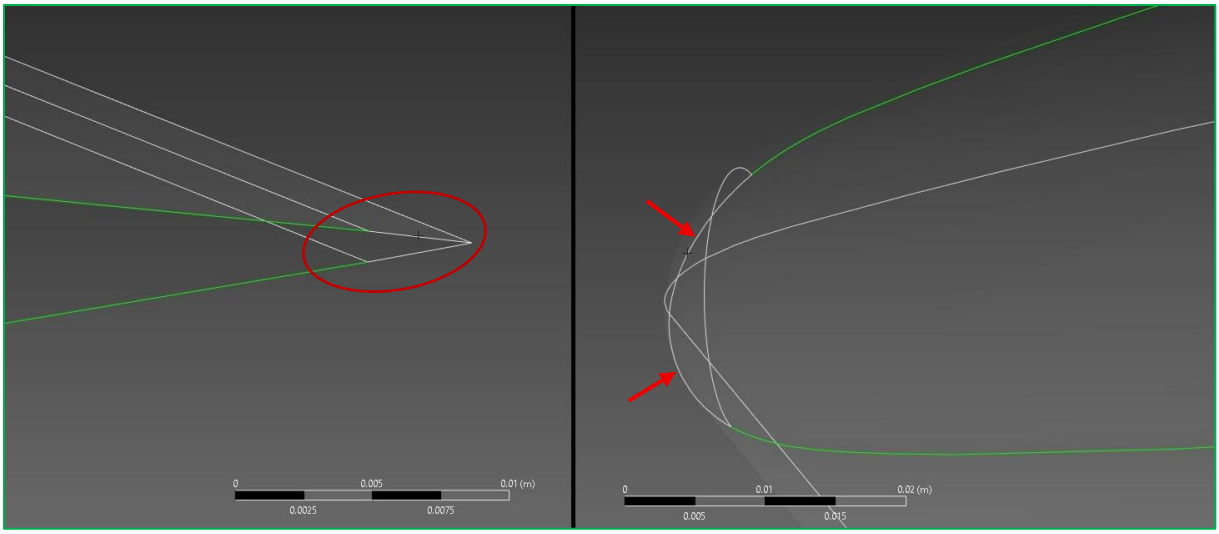


Figure 3.16: These edges can easily be missed, pay close attention to LE and TE of the Wing when selecting

3.2.c Spanwise Edge Sizing

Mesh > Insert > Sizing

With the *Edge* selection tool (see the icon at the top ribbon, or Ctrl+E), select the following edges, and then apply the settings shown (**Figure 3.17**). Again, we are using an arbitrary sizing for this at 0.004m. Please note, there are multiple edges across the span around the back of the wing, and make sure you pick the trailing edge, the single line spanning from root to tip. Once everything looks clean, hit update.

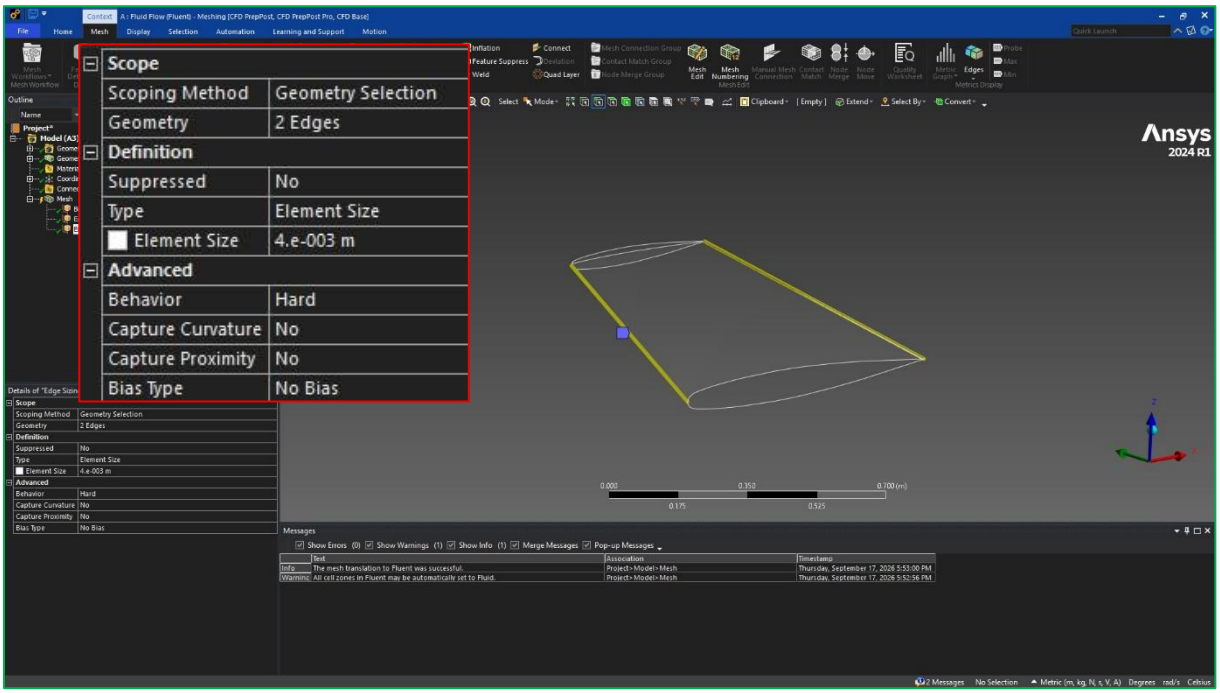


Figure 3.17: Sizing of spanwise edges of the wing

3.2.d Wing Surface Sizing

Before we get to sizing for the wing, this is a good time to define our first 'Named Selection', essential components of the fluid domain that carries significance throughout the simulation (and beyond). Think of them as IDs. Define them once, and they can be identified (to work with) throughout the entire project, even in CFD-post.

Make sure Face selection tool is enabled (Ctrl+F), and holding Ctrl key, select all the faces (**Figure 3.18**) that represents the wing (or an adiabatic wall, in the context of fluid mechanics). There will be a total of 10 faces – marked in the figure. Note, some of the faces appear in both pictures, but they are the same faces, in two different frames.

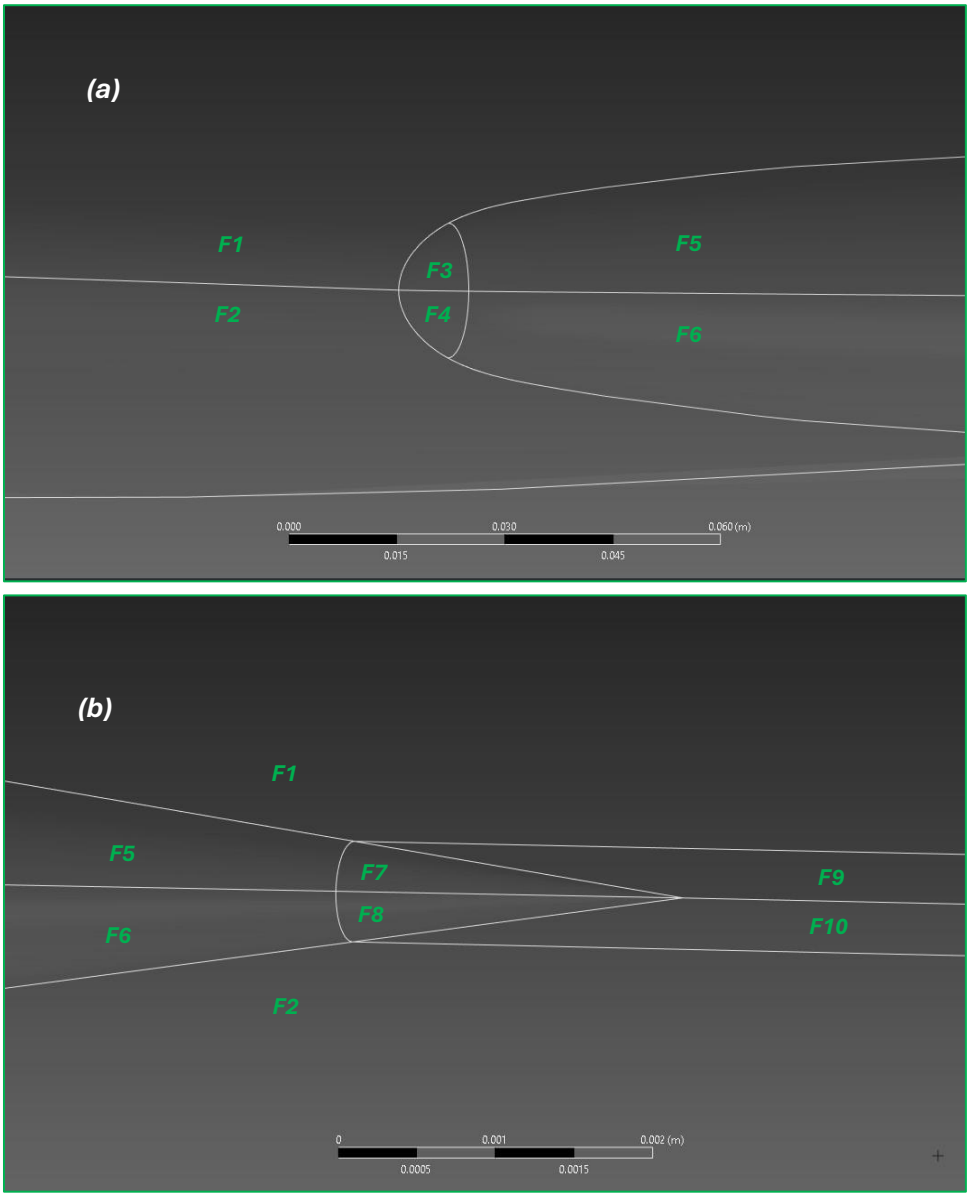


Figure 3.18: All faces of wing (a) zoomed in at tip LE, (b) zoomed in at tip TE

Once all the faces are selected, hit N to create a *Named Selection*. Lets name it 'airfoil-surface'. Hit OK. If you selected all the surfaces, you can click on the + sign beside Named Selection on the feature tree, click on the airfoil-surface, and should be seeing it has 10 faces, with some additional values (**Figure 3.19**). We can now add sizing to it.

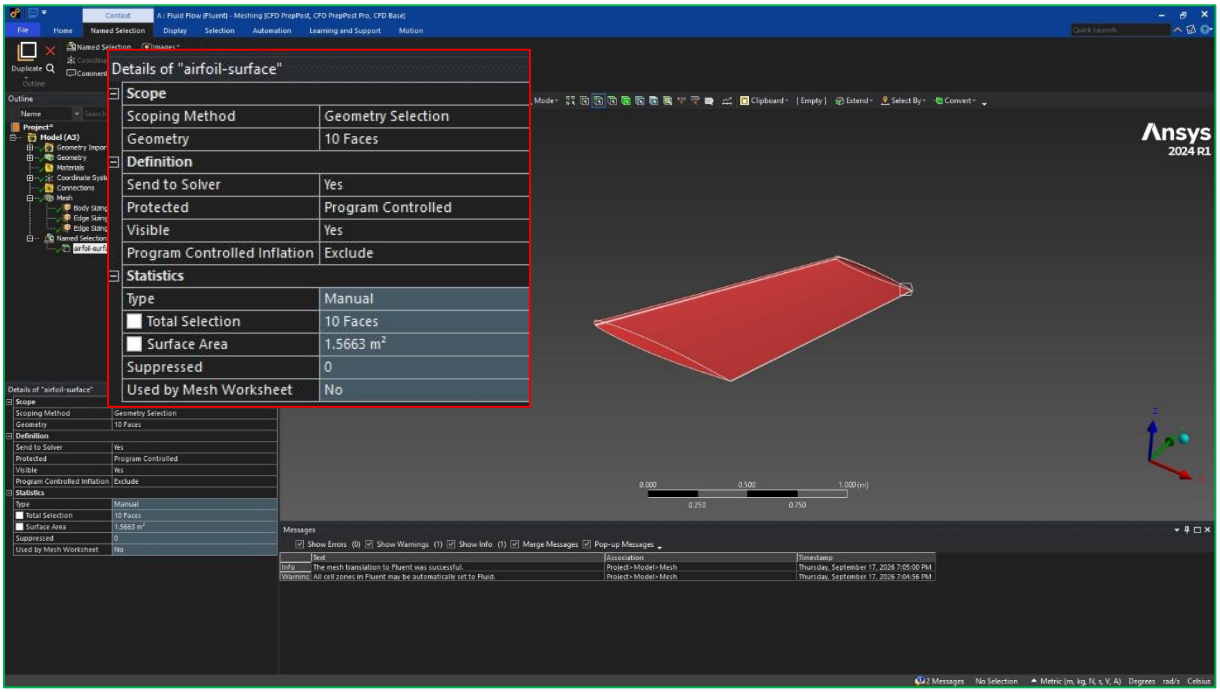


Figure 3.19: Named Selection showing basic properties of airfoil-surface

Mesh > Insert > Sizing

In scoping method, use the Named Selection option instead of the default, and select the 'airfoil-surface' from the dropdown menu. Again, using an arbitrary value of 0.02m for this sizing the airfoil surface. Keep all the other settings as is (**Figure 3.20**).

Hit update.

> *Note – In the named selection, I used 'airfoil-surface' as the name. While this is a wing analysis and not 2D airfoil surface, the naming doesn't matter much. Force of habit, if you will. When I first learned about this, I used 'airfoil-surface'. And as mentioned, since this is an ID for the entire pipeline of work, the naming convention sticks. You can absolutely name it 'wing-surface', the difference will be the ID itself later down the line – the method remains the same regardless of what you name it.*

What's in a name, eh!? Sorry for the slight confusion!

(We are really close to finishing the meshing – I know it's been a long, long journey but trust me it's all going to be worth it… depending on who you are asking of course!)

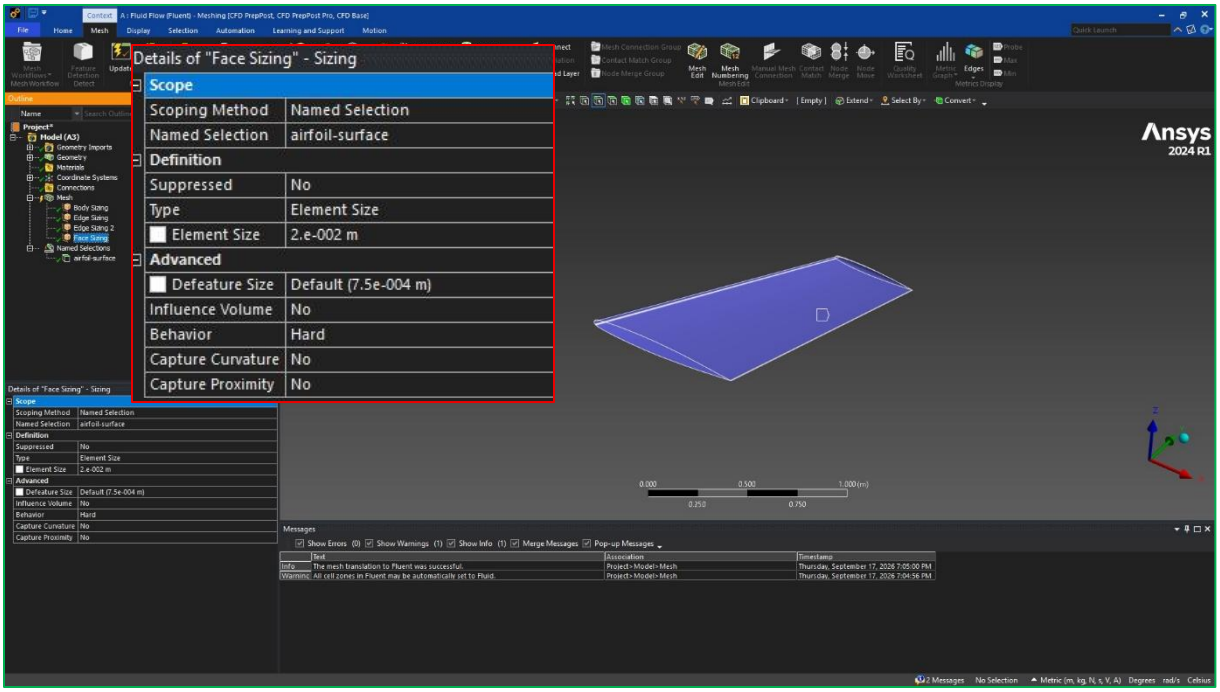


Figure 3.20: Sizing method for Wing Surface

3.2.e Inflation Layer

While this section is very crucial, mathematically – the whole extent of the logic is beyond the scope of this instruction manual. Before setting the inflation layer (which is essentially defining the 'no-slip' condition over the wing), you must do some calculations on the assumed y+ value. In future, I wish to add the mathematics behind estimating the thickness (a key settings in this part, more below). But for now, like the rest of the meshing method, we are going to use an arbitrary value.

Anyways, I am gonna go straight to it. It's way too long at this point!

Mesh > Insert > Inflation

Select the fluid domain in the *Scoping Method*. And for the *Boundary Scoping Method*, select *Named Selection*, and then select the 'airfoil-surface' we defined earlier. Then in the *Inflation Option*, select *First Layer Thickness*. The arbitrary value here is 9.25e-6. Select the Maximum Layers to be 15, with a growth rate of 1.2 (**Figure 3.21**). The whole point of selecting these settings is to make sure we get a y+ value < 1, however anything below 5 is respectable (at least for initial learning) for the turbulence model we will be using for a properly resolved boundary layer. We can check what our wall y+ is in CFD-post.

Does that much information make your head spin? Good. That makes two of us!

(not really, but I said that for the dramatic effect!)

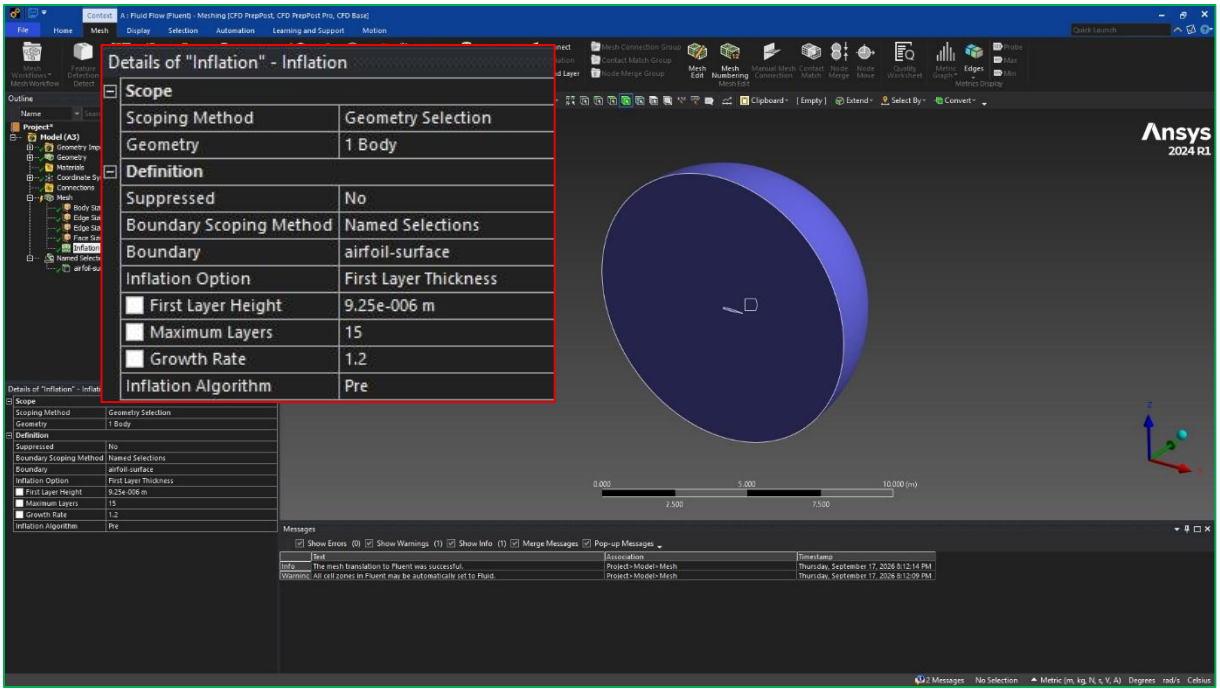


Figure 3.21: Inflation Layer Setup

Hit update. After the mesh is generated, hit Ctrl+F (Face Selection tool), and click the flat plane of the fluid domain. Hit N, and name it 'symmetry'. Then, select the curved surface, hit N, and name it 'farfield' (**Figure 3.22**). Hit update. After the update is done (shouldn't take more than 2/3 seconds anyways), check if you have a 'tick' mark beside Mesh in your workbench. If you see the green tick mark, you can close the meshing software safely.

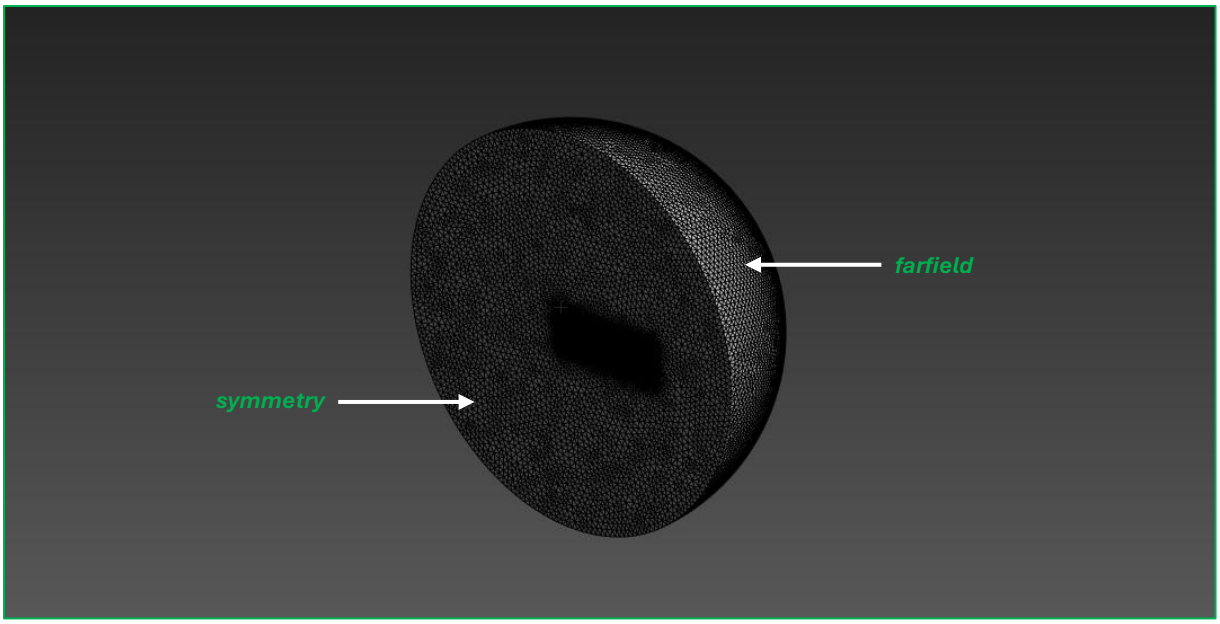


Figure 3.22: Named Selections for Components of the Fluid Domain

Yes. We are done meshing!

Part 4

FLUENT SOLVER (FLUID SETUP)

Let us set up the 'digital wind tunnel', shall we?

After you finished meshing and checking that the green tick mark is there for *Mesh*, given that you closed the Meshing software and are in the workbench, it's a good practice to save the project. Once you do so, double click *Setup*. In the pop-up window, select Double Precision and choose how many solver process you want to use. This is obviously workstation-specific. For example, my pc has 8 cores and I am selecting 6 processes (**Figure 4.1**), giving the poor old workstation a breathing room of 2 cores!

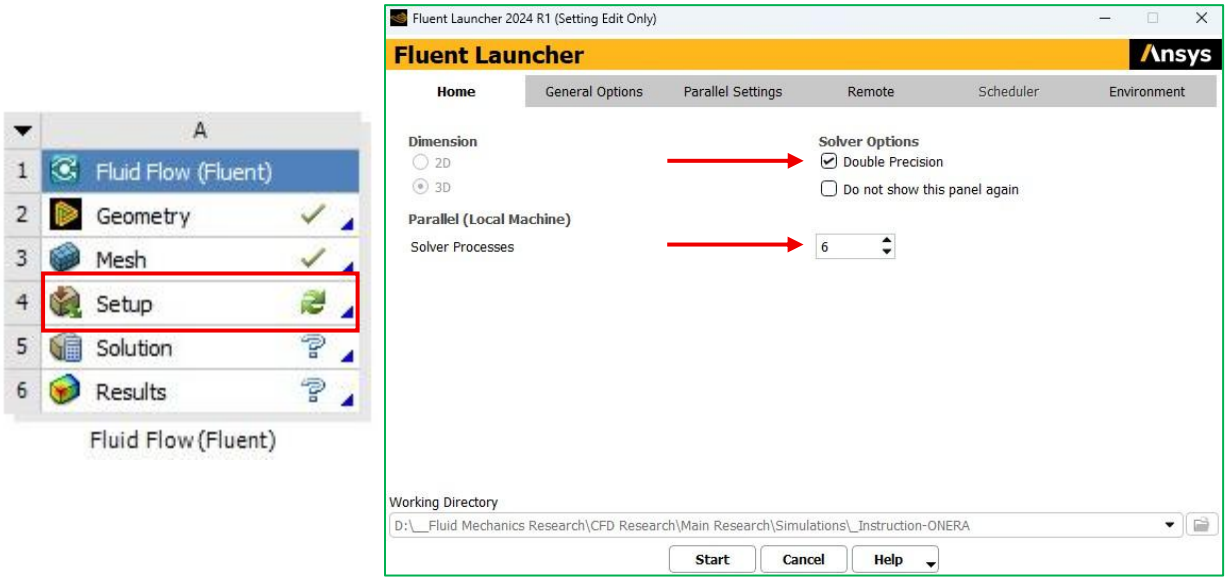


Figure 4.1: Fluent Settings

4.1 General Settings and Fluid Setup

We will be using a *Pressure-Based* solver, with the Spalart-Allmaras turbulence model. All the settings in *General* should be default; however, it is still worth checking your default settings looks exactly as is (**Figure 4.2**). In the *Models* section, double click on *Viscous* and select the *Spalart-Allmaras (1 eqn)* option (by default, this is selected as the SST k-omega). The *Spalart-Allmaras Production* should be *Strain/Vorticity Based*. Leave all the other settings as default in this window, since we don't need to worry too much about them for our study.

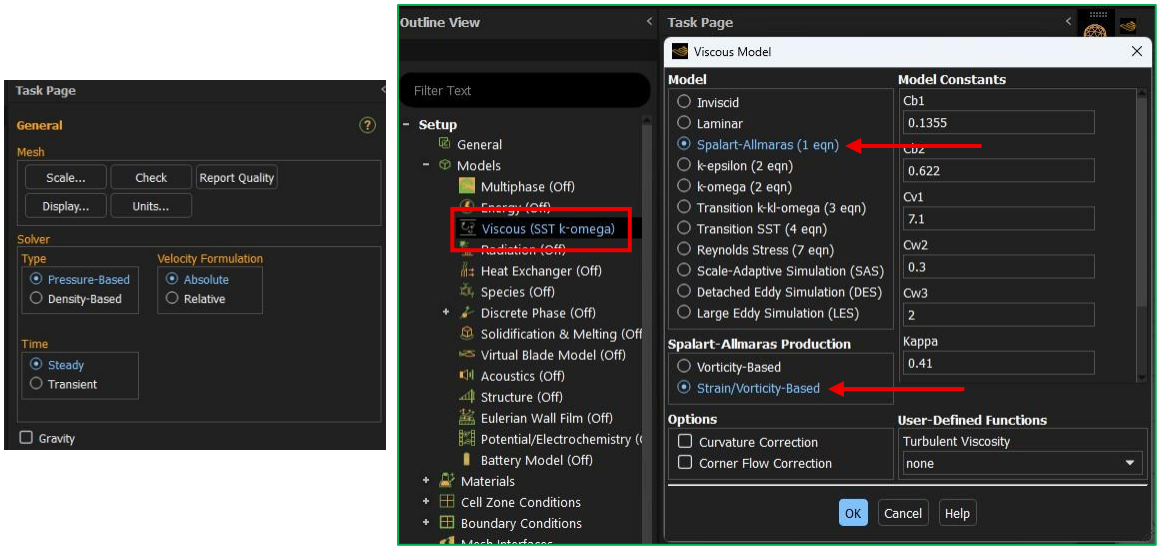


Figure 4.2: Model Setup in Fluent

Time to define the fluid. Since we are working with compressible flow, use the settings shown below (**Figure 4.3**) in *Materials > Fluid > air*.

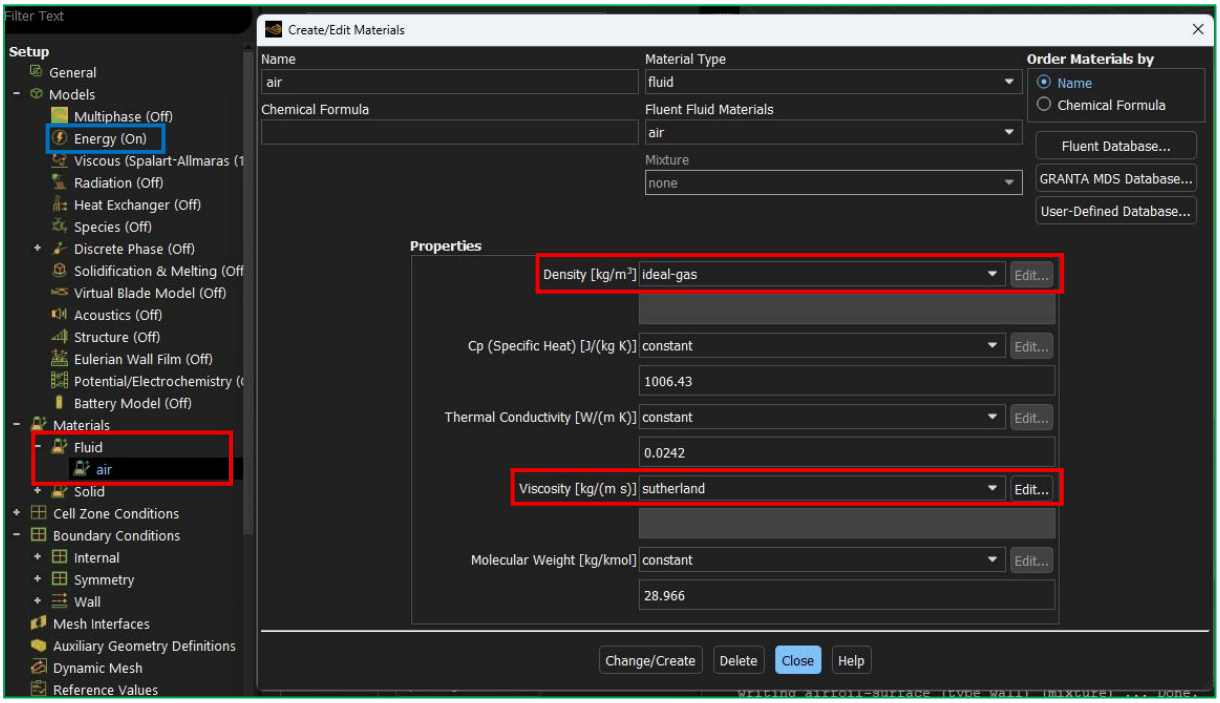


Figure 4.3: Air Properties Settings

Once you select the Sutherland viscosity option, you will be prompted to define the coefficients for the Sutherland Law (**Figure 4.4**). Note, you will not see *Energy (On)* by default (marked in blue). But once you define the air with Sutherland viscous model, it will be automatically turned on.

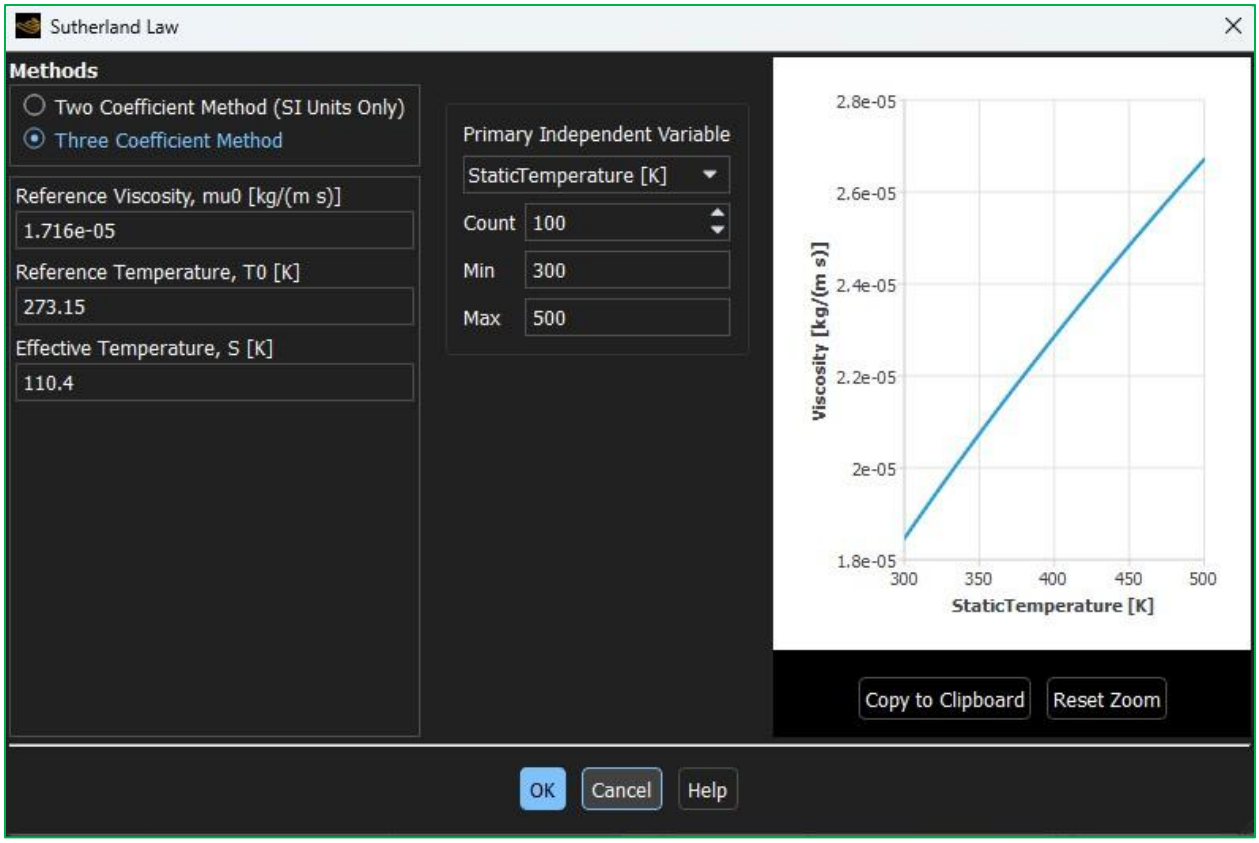


Figure 4.4: Sutherland Law Coefficients

Here is a checklist of the settings –

- Density ideal-gas
- Viscosity Sutherland
- Sutherland Law
  - *Reference Viscosity: 1.716e-05*
  - *Reference Temperature: 273.15*
  - *Effective Temperature: 110.4*

In Sutherland Law window, hit *OK*. Then hit *Change/Create* in Create/Edit Material window. These settings should turn on Energy Equation (see feature tree). The fluid is defined.

4.2 Operating and Boundary Conditions

Set the operating pressure to 0 by double clicking *Boundary Conditions* and clicking *Operating Conditions…* (**Figure 4.5**).

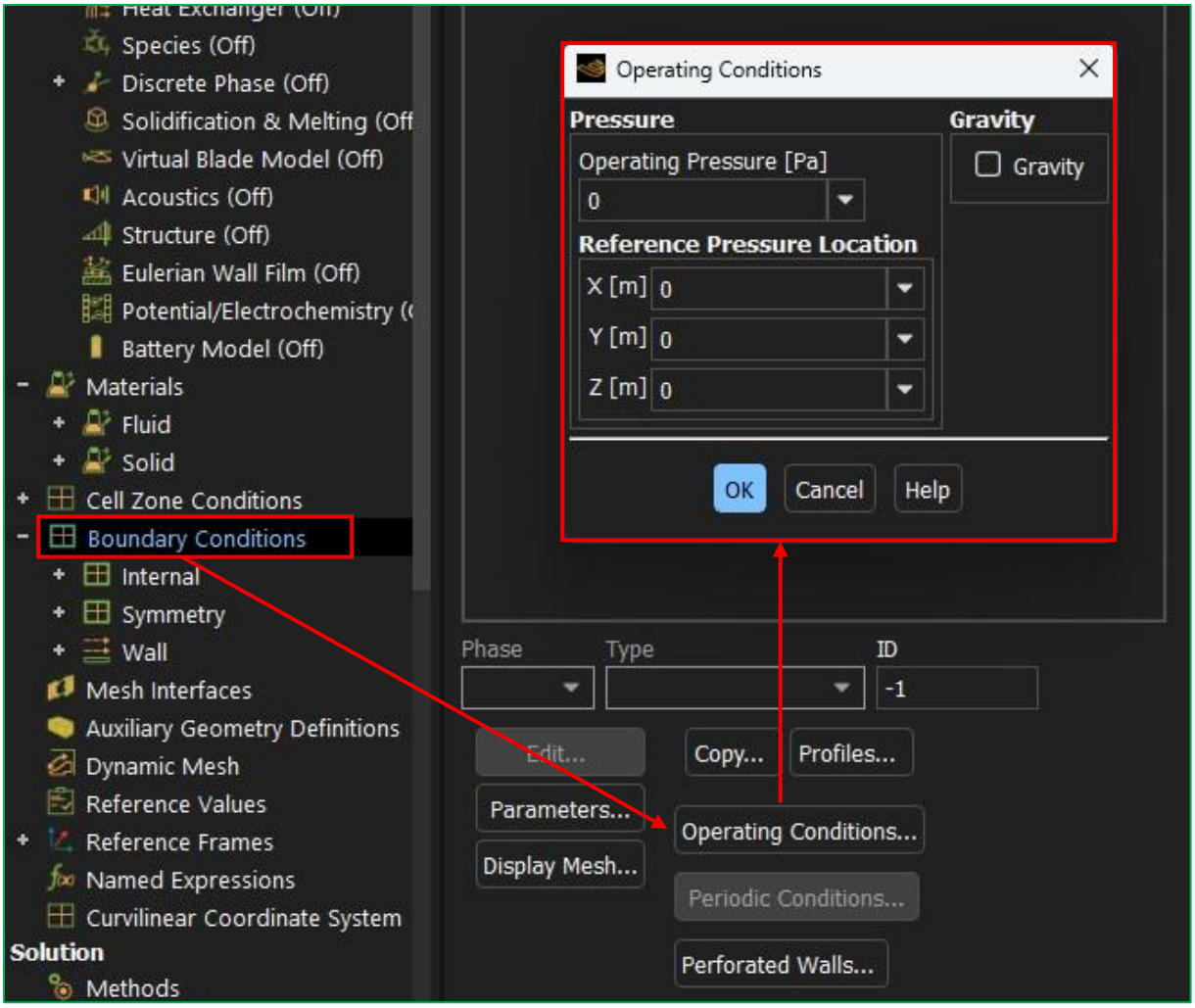


Figure 4.5: Operating Conditions Setup

In *Boundary Conditions > Wall*, right click on *farfield* and under *Type*, select *pressure-far-field*. This is where you define the velocity (magnitude, direction) of the flow. In our case, the values are Mach 0.84 and 3.06° Angle of Attack. Additionally, we will set a Gauge Pressure of 97913.3 Pa. These values are case specific of course. For example, the gauge pressure can be calculated from the alternative form of ideal gas law –

$$P = \rho R T$$

And we will be using the trigonometric component values (for X-axis, and Z-axis) for the fluid direction. Since our model is setup in such way that the wing depth is in the Y-axis, the Z-axis becomes the vertical direction for the velocity, while X-axis stays the horizontal direction. The values shown below (**Figure 4.6**) represents components for 3.06° Angle of Attack. Also, select *Modified Turbulent Viscosity* under *Specification Method* in the same window. The value for that to be 4.85488E-5. In the *Thermal* tab, make sure *Temperature* is set to 300K as well. Hit *Apply* and we are done setting up the fluid!

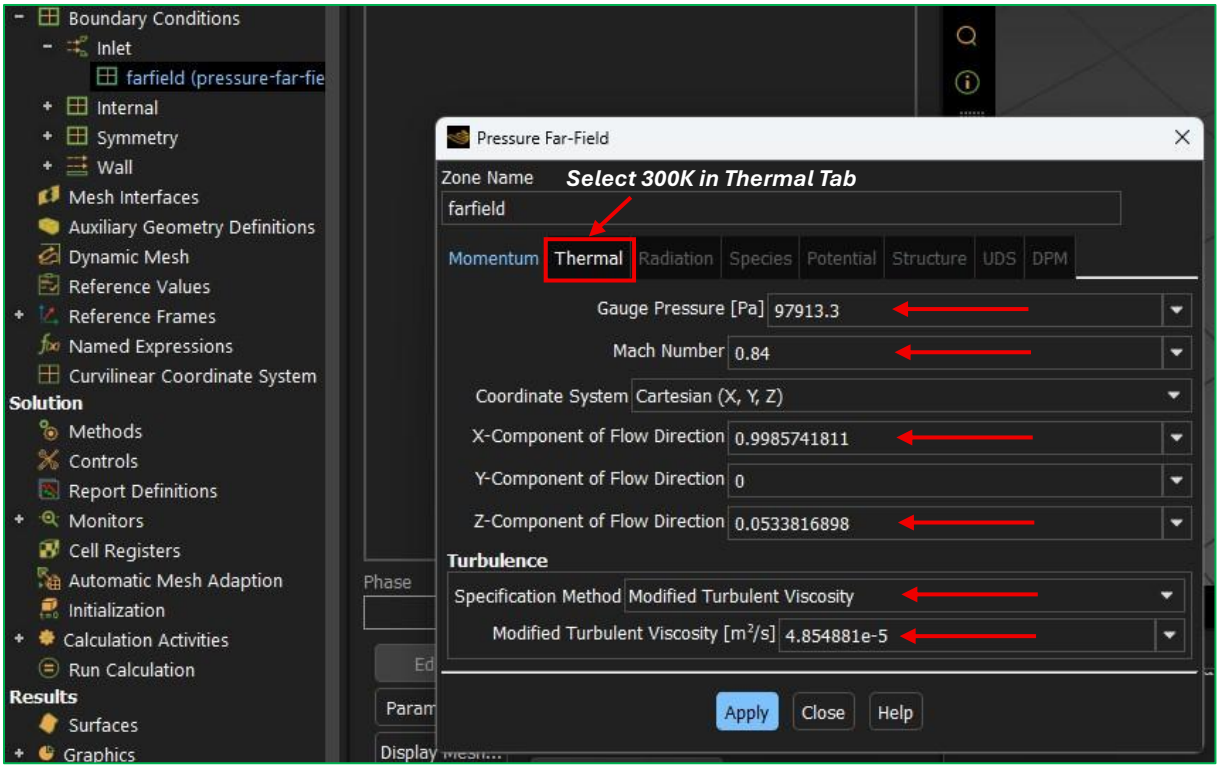


Figure 4.5: Operating Conditions Setup

Here is a quick recap of all the values you need in setting up the fluid for the farfield boundary conditions (inlet) –

Parameter	Value
Gauge Pressure [Pa}	97913.3
Mach Number	0.84
X-Component of Flow Direction	0.9985741811 [cos(3.06)]
Y-Component of Flow Direction	0
Z-Component of Flow Direction	0.0533816898 [sin(3.06)]
Specification Method	Modified Turbulent Viscosity
Modified Turbulent Viscosity [m^2/s]	4.854881e-5

Next, double click on the *Reference Values*, and select *farfield* in the *Compute from* option. The *Density [kg/m^3]* and *Viscosity [kg/(m s)]* should be correctly calculated by Fluent, and you should see the values as such, shown below (**Figure: 4.6**). Also, setup the Reference Values for Area and Length. Again, this is a case-by-case option and the values represents the ONERA M6 geometry we are working with.

Parameter	Value
Area [m^2]	0.76027996
Length [m]	0.81

You can find the value for the area from the CAD files (evaluate). And the length is for root chord – the specific characteristics length used in the study we are referencing.

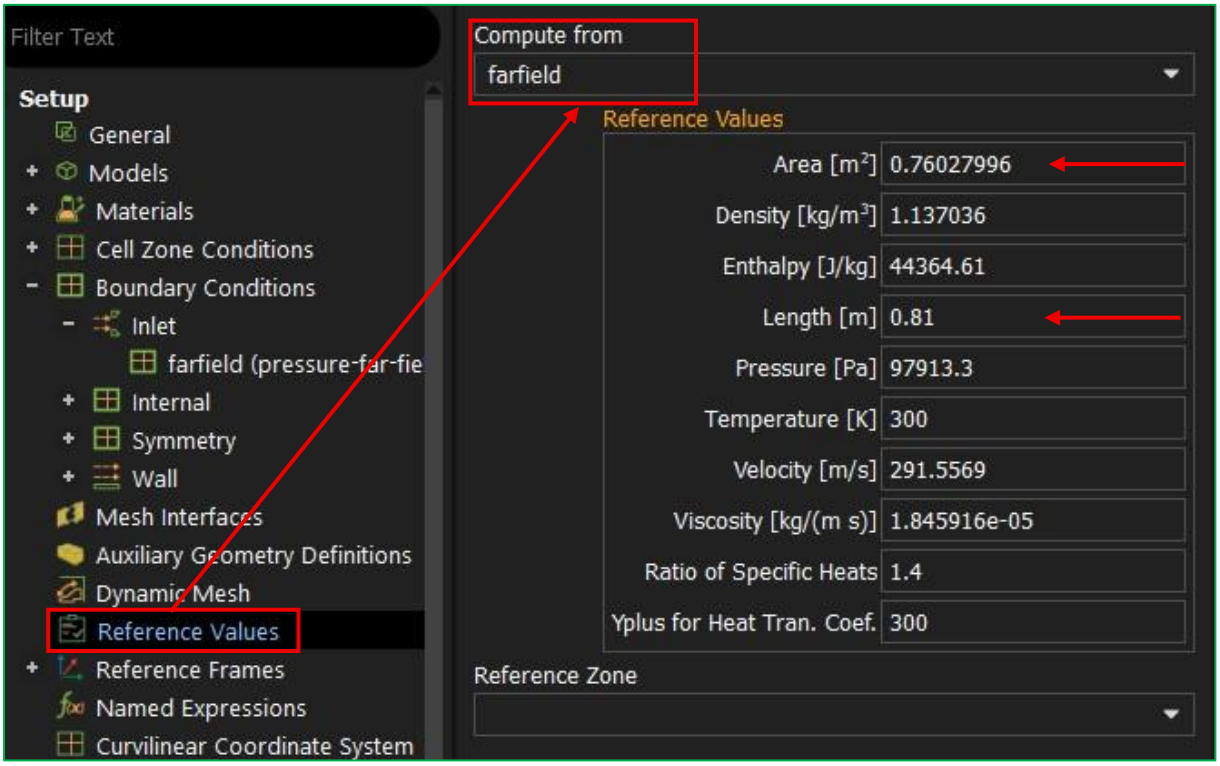


Figure 4.5: Operating Conditions Setup

4.3 Coefficient Report Setup (Report Definition)

One last thing before we hit run –

We will define the primary coefficients (Drag and Lift) for our results, that we can use to measure against the published data. This is where the earlier component values (for Angle of Attack manipulation) come back, and we don't need to compute anything new. The convention goes like this (consult the previous table for the values) –

Parameter	Value
Drag Force Vector [X]	(X-Component of Flow Direction)
Drag Force Vector [Y]	0
Drag Force Vector [Z]	(Z-Component of Flow Direction)
Lift Force Vector [X]	- (Z-Component of Flow Direction)
Lift Force Vector [Y]	0
Lift Force Vector [Z]	(X-Component of Flow Direction)
Zones (for both Drag and Lift)	airfoil-surface

Right click on *Report Definition*, then navigate to *New > Force Report > Drag*. In the pop-up window, input the values as stated in the table above. Make sure to rename (if you want), and put the tick mark beside Print to Console. If all looks good, hit OK. Do the same thing for lift as well (*New > Force Report > Lift)*.

Make sure all the values (and options) are as is shown in **Figure 4.6**.

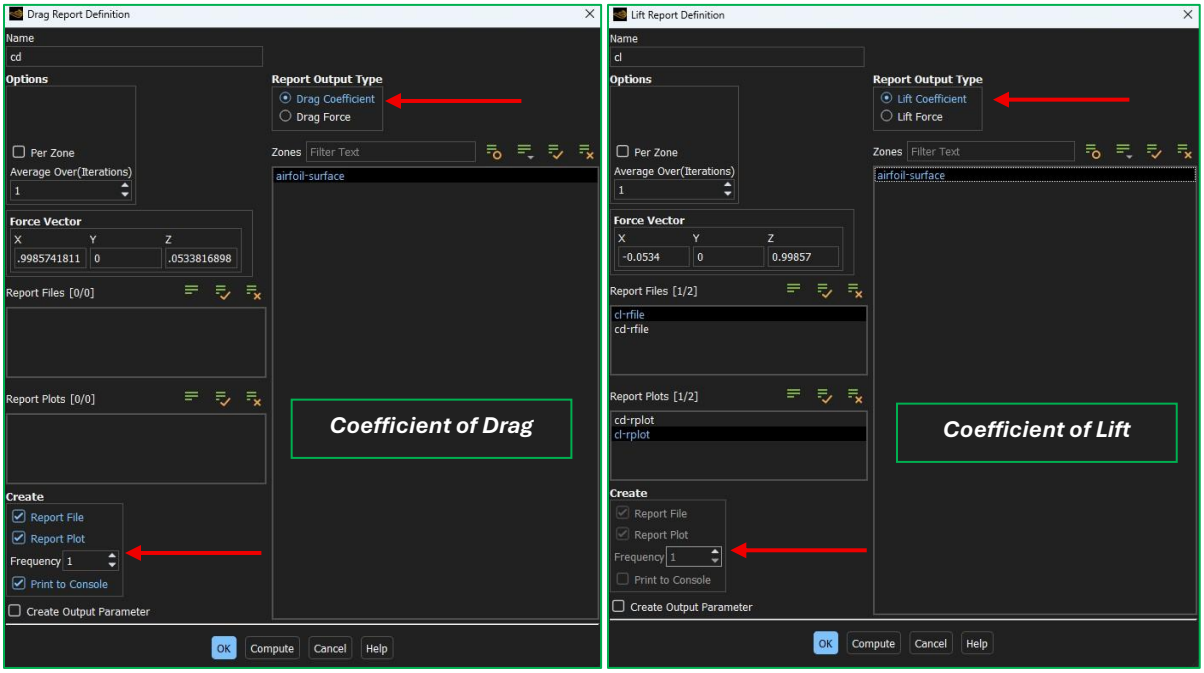


Figure 4.6: Report Definition Setup

Now, expand *Monitors*, double click on *Residuals* and make sure the *Absolute Criteria* is set to *1e-6* for all the equations (**Figure 4.7**). This is a conservative approach, but smaller values can be entertained if required.

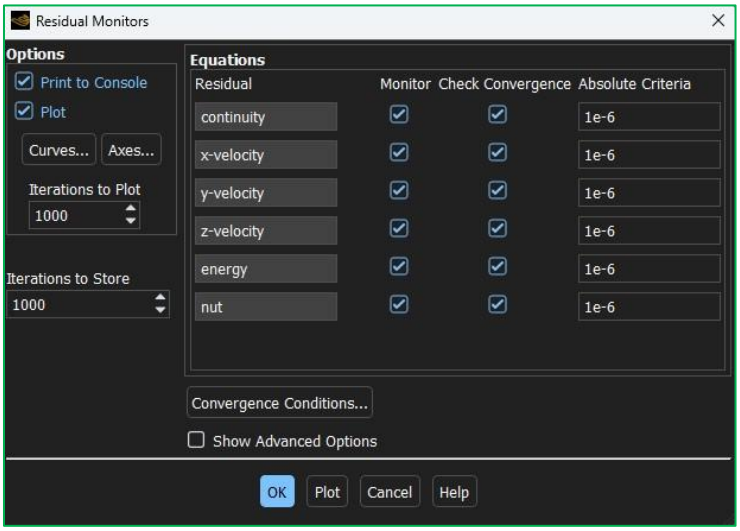


Figure 4.7: Residual Criteria

And just like that, our solution is ready to be calculated!

Go to *Initialization* under *Solution*, select *Standard Initialization* method, select *farfield* for *Compute from*, cross-check all the values are correct, and hit Initialize (**Figure 4.8**). Please note, Hybrid Initialization is also acceptable.

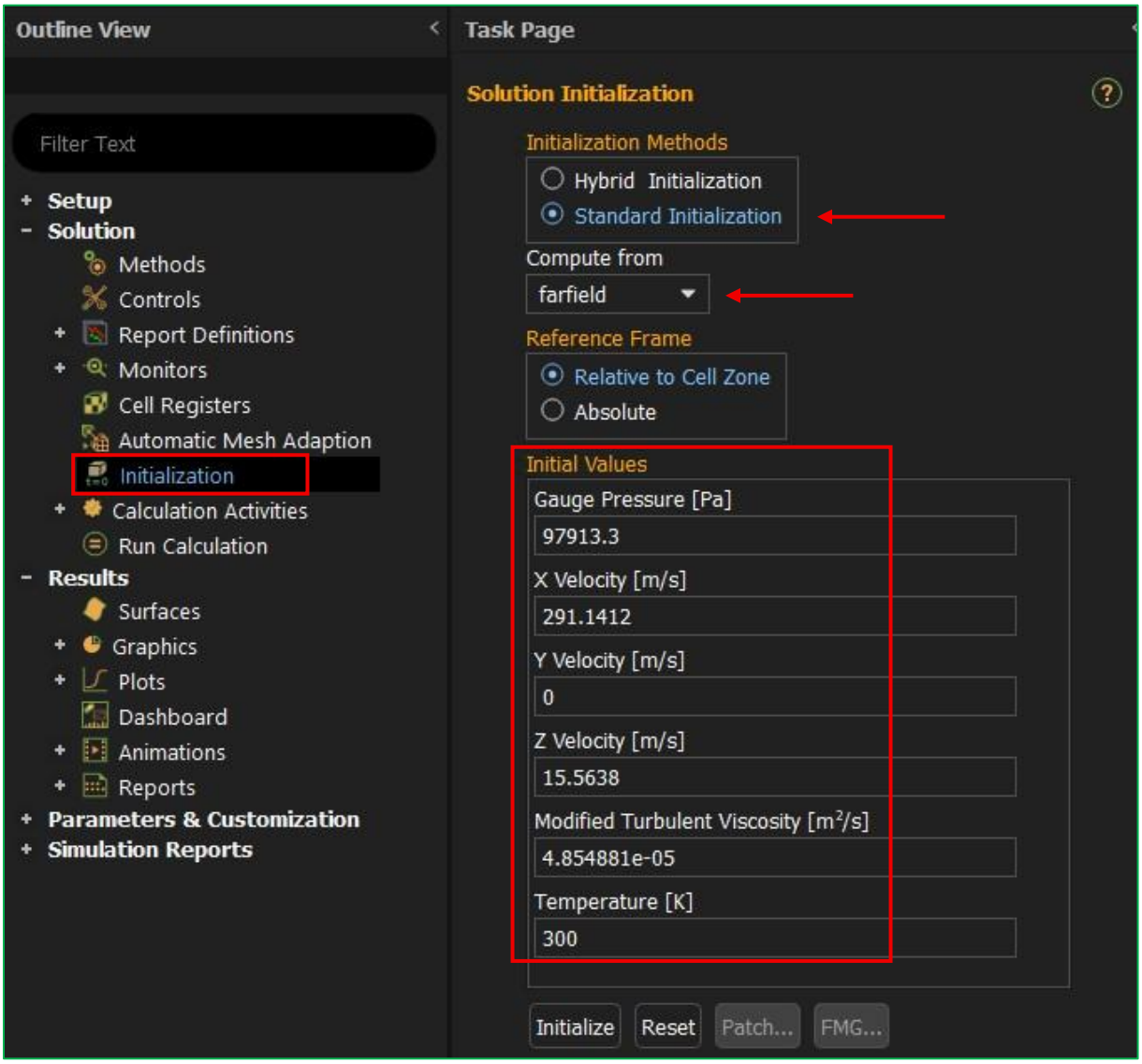


Figure 4.8: Solution Initialization

Once the solution is initialized, double click on *Run Calculation*. We will be using 500 *Number of Iterations* for this run (**Figure 4.9**). Then select *Data File Quantities…* and make sure to include the following values to be included in the solution (**Figure 4.10**) –

- Pressure Coefficient
- Mach Number
- Wall Yplus

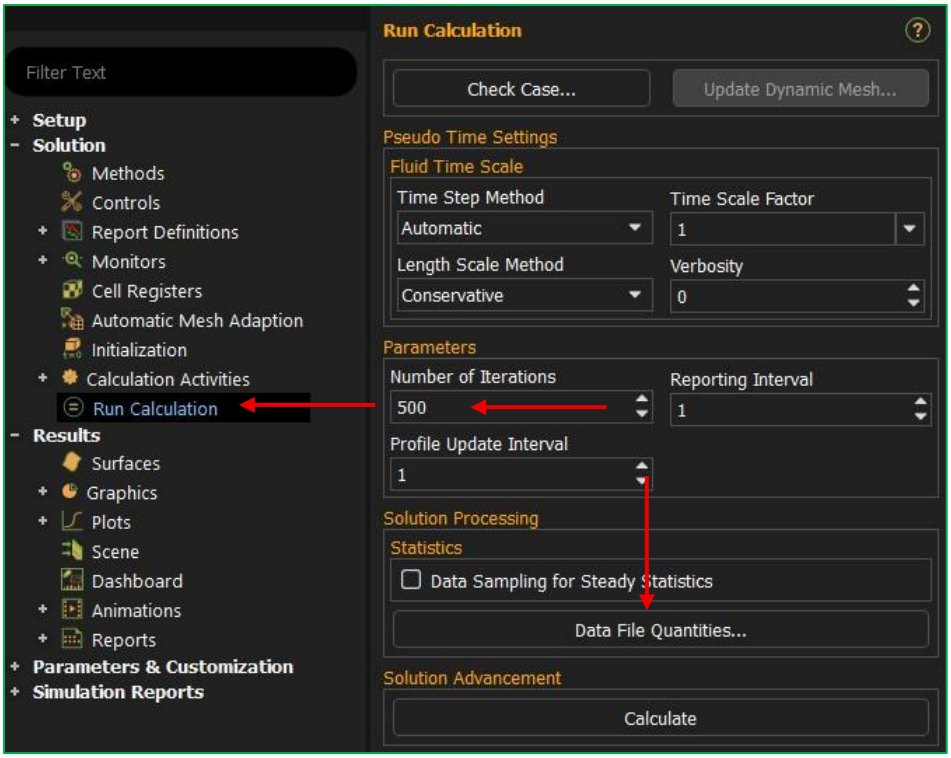


Figure 4.9: Run Calculation

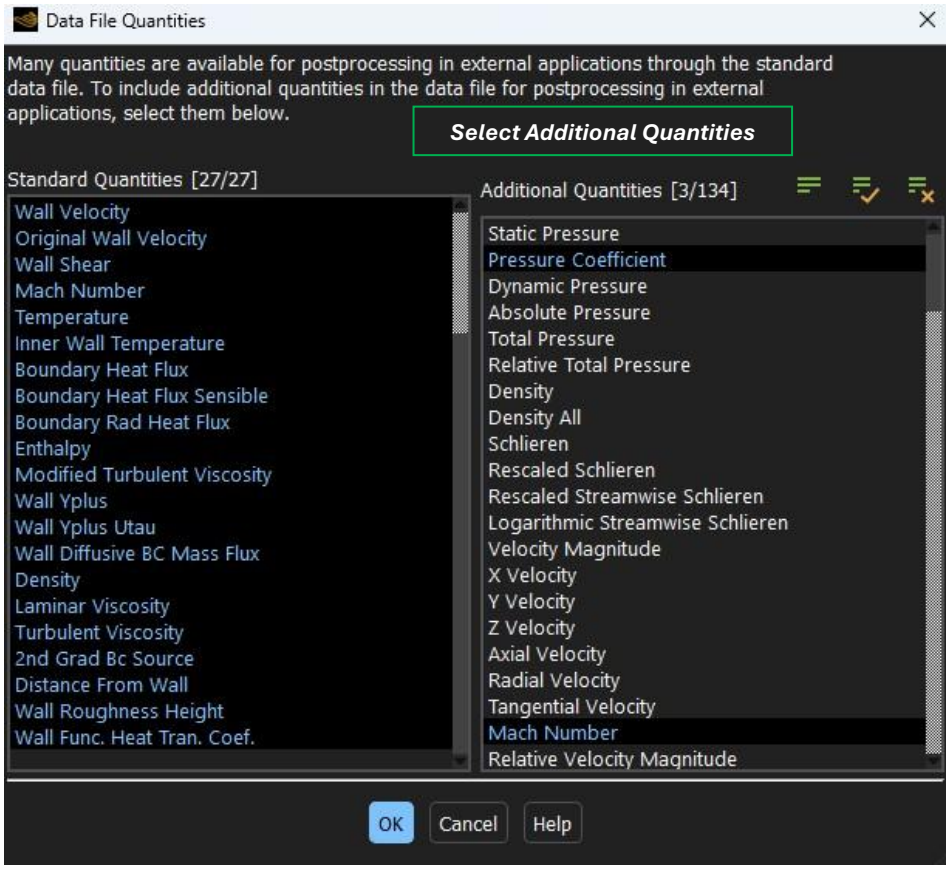


Figure 4.10: Selecting Additional Quantities

Hit OK Yes, we are finally done! Glad you made it to this point!

All is left to do now – hit the Calculate button in the Run Calculation task page and let fluent do its thing!

While your computer is crunching numbers, it's a good time to grab a cup of coffee or something. We will discuss the coefficient values as the 'base' for the validity of our work. And that will be the only result discussion in this instruction.

Once the solution is run, go to *Files* and select *Close Fluent*. In the workbench, save the project first, and we can start with the initial results.

Part 5

INITIAL RESULTS DISCUSSION

Now, we can extract some of the data to check against published results.

Double click on Solution (**Figure 5.1**).

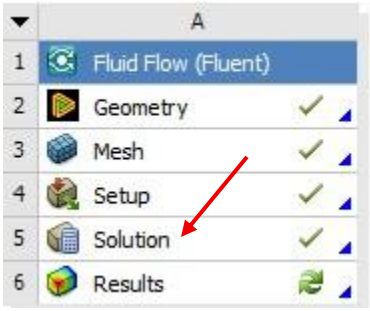


Figure 5.1: Solution Tab on Project

Once Fluent loads with our solution data, expand *Report Definition*, right click on *cl*, and hit compute. Do the same for *cd* as well. You should have something closer to the following values, if the whole project was done right –

Parameter	Value
Cl	0.25697196
Cd	0.016899552

Looking at the published data, both of them seem to match pretty close (**Figure 5.2**). Despite our solution underpredicting the lift force slightly, it is still a good place to land on – specifically given our mesh was not too dense (kept in such way by design to reduce compute time). As you make finer adjustments to the mesh, the lift data should get closer to the published results.

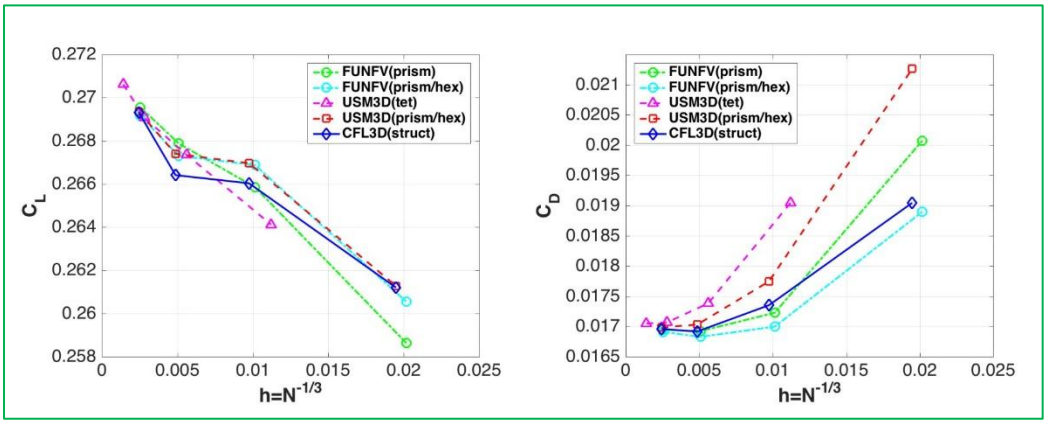


Figure 5.2: Published data for ONERA M6 (Diskin et al., 2018)

In addition to some of the coefficient results, we can also plot some contours in Fluent. But do note, these are only for initial validation. The standard practice is to use CFD-post for result generation, especially if you wish to publish.

5.1 Mach Contour at Symmetry Plane

In *Results > Graphics*, locate *Contours* and right click to select *New...* In the window, select *Velocity...* under Contours of, and then select *Mach Number* from the following dropdown menu. Select *symmetry* on *Surfaces*, keep all the other options default and hit Save/Display.

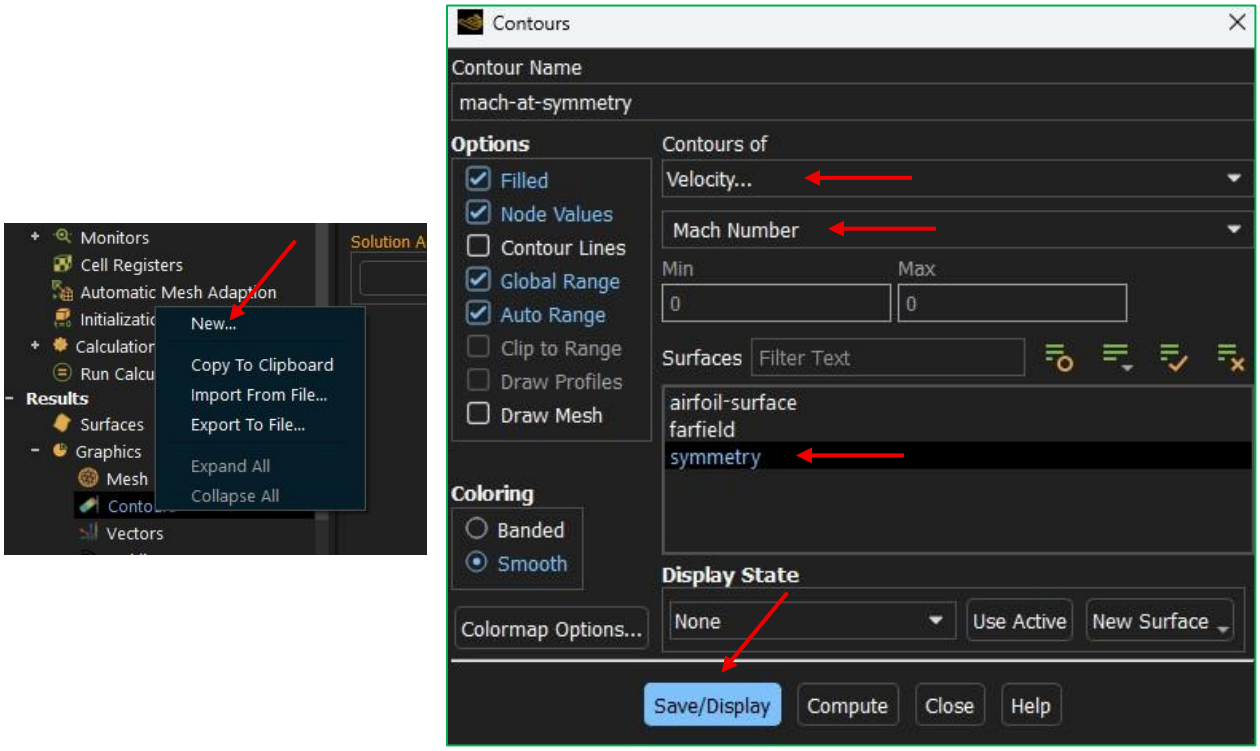


Figure 5.3: Plotting Mach Contour

5.2 Pressure Distribution over Wing

Following the same procedure as 5.1, you can select *Pressure...* for *Contours of*, and then select *Pressure Coefficient* on the following dropdown menu. Select *airfoil-surface* this time and hit Save/Display.

You should be able to see the shockwave line over the upper surface of the wing.

And that is about it. I added a couple of pictures in *Appendix A* to showcase the contour plots, if you want to check it out.

Concluding Remarks

We are done. Finally!

Congratulations on finishing this long, long instruction manual. But if done right, you have a working CFD-analysis of ONERA M6, that can be used to extract valuable data in the post-processing. I hope that counts as the reward for your patience!

I wish to develop a separate solution manual for CFD-post, since that discussion is beyond the scope of the current manual.

(it is already too long!)

I hope this manual helps you learn 3D CFD. Of course, the full extent of the project is beyond anything I can write about, since CFD itself is a vast, active domain of research. Academics and practitioners are constantly finding new ways to do numerical analysis, and contributing to the field at large. If you are serious about computational physics, I highly suggest you to keep an eye on current literature, specifically when it comes to new, novel ways to do computational fluid dynamic simulations. Some of the work out there is fascinating, and I don't think I will be able to stop talking about it if I start.

So, I am not going to start.

(also, I hardly understand some of the unique techniques the authors are employing but you did not hear that from me!)

Best of luck to your learning journey to computational fluid dynamics, and beyond!

Omar Saif
CFD Enthusiast
October 6, 2026

Appendix A

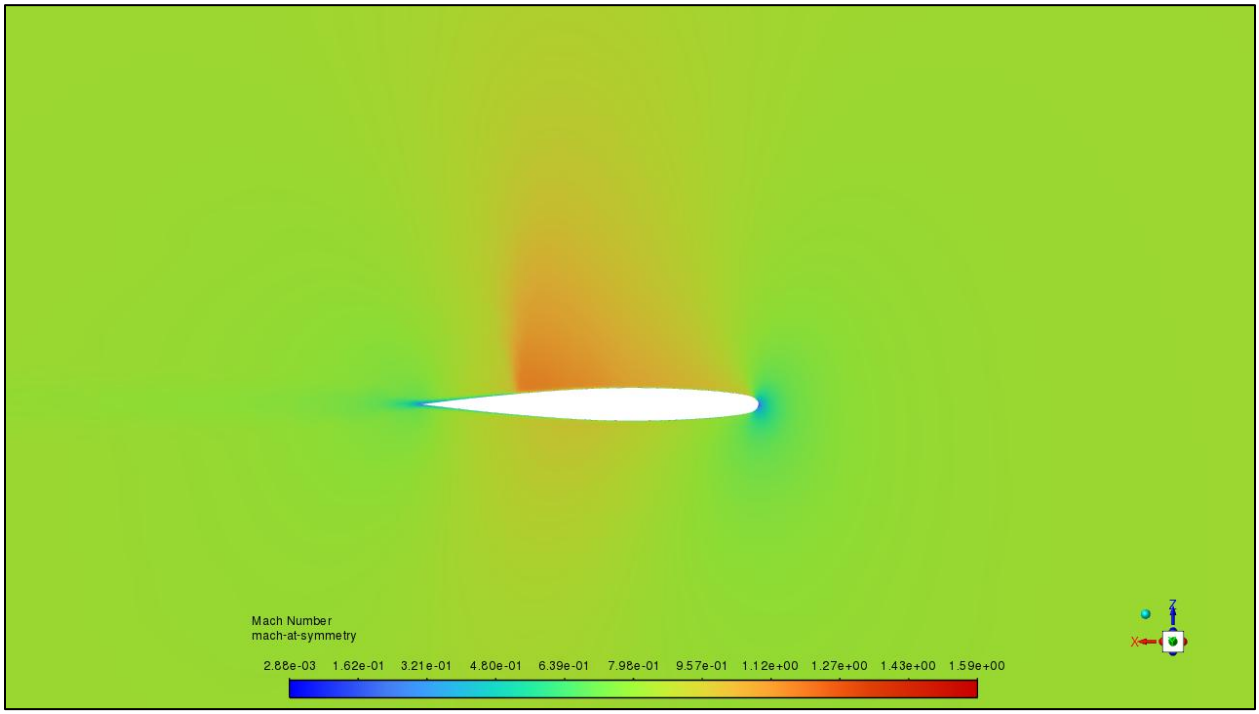


Figure A-1: Mach Contour at Symmetry Plane

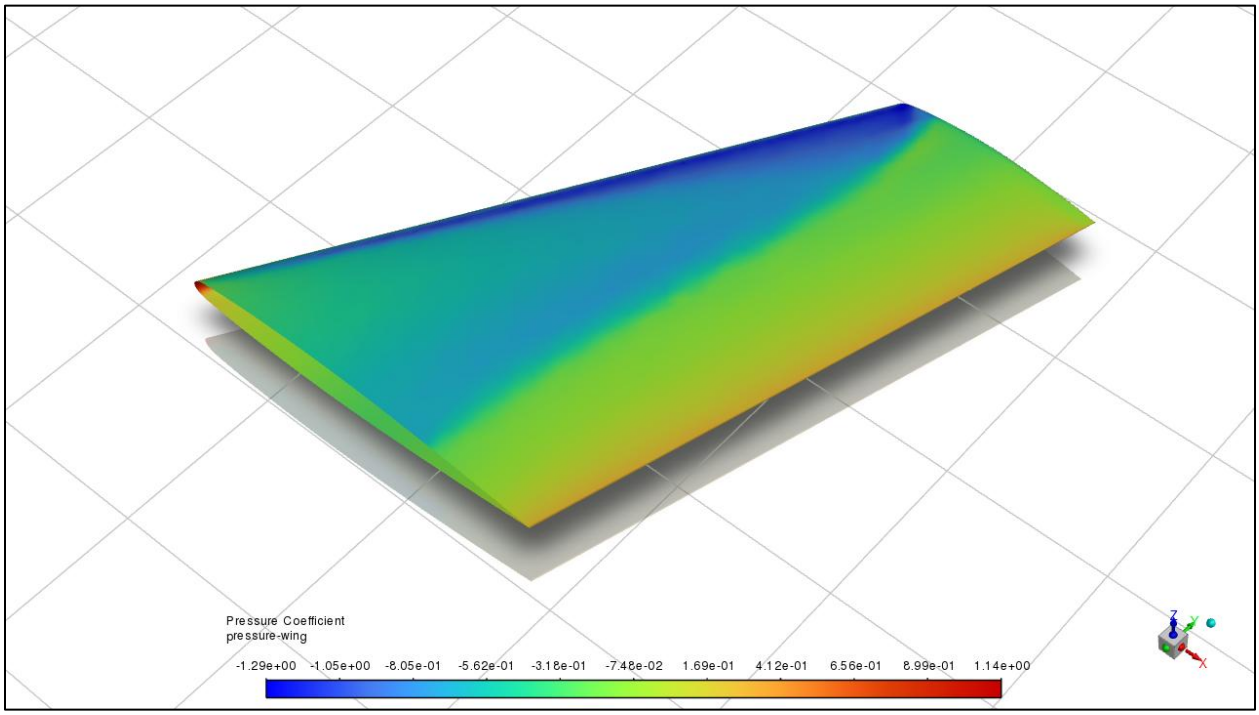


Figure A-2: Pressure Distribution over Wing Surface